

Universidade de São Paulo
Faculdade de Economia, Administração, Contabilidade e Atuária
Departamento de Economia

RAFAEL PIRES DE OLIVEIRA

**ESSAYS ON REGIONAL ECONOMICS AND
LABOR MARKET**

São Paulo
2025

Rafael Pires de Oliveira

**Essays on Regional Economics and
Labor Market**

Monograph submitted to the School of Economics at the University of São Paulo as part of the requirements for the degree of Bachelor in Economics.

University of São Paulo

Orientador: Prof. Dr. Carlos Roberto Azzoni

São Paulo
2025

To my parents.

Agradecimentos

Compiling a list of acknowledgments is difficult, but it is an extremely necessary task. I am reminded of a folk tale: a baker suspected that the kilogram of butter he was buying from a farmer was “shrinking.” The baker then took the case to the judge, who asked how the farmer weighed the butter. The farmer’s answer was simple: “I have a scale, but I don’t have a standard weight. I put the bread the baker gives me on one side and the butter on the other. I deliver what I receive.” This lesson stays with me. As I write this work, I realize that much of what I carry came from others.

To my grandparents, who came from the Northeastern hinterlands, with courage in their hearts and dreams in their luggage. In the midst of hyperinflation, when many retreated, they ventured and resisted with dignity. With little, they did much. My grandfather taught me to do the math when I barely knew what economics was; my grandmother cried when she learned that this was what I wanted to study. They showed that no number makes sense without memory and without people.

To my parents, who have always been by my side. They shared the burden of the day and of doubt, working tirelessly so that I could have what they never had: the peace of mind to make mistakes. If I write this today, it is because they never stopped believing in me, even when I doubted myself.

To the brothers who made this journey something greater: João Pedro (*Jão da Massa*) and André Casalta (*André Balada*). To JP, Vini, FG, Luiz, Eric, Marrey, Moutta, Leozinho, Thiago, Otto, Daniboy, Laura, Mariane, Giovanna, Caju, and so many others, for the partnership at FEA and in life. To those who arrived later and became fundamental, like Dani Minson, with her exceptional comfort. To Bruno Mike and João Victor (as well as everyone from *Nosso Date*), for the company and tireless shoulders to lean on. To Thom, one of my best friends, even if I have made the mistake of, occasionally, not realizing it.

To the one who, surpassing any praise, occupied the place in my heart that only the most intense and beautiful love could occupy.

To all the professors at FEA, especially Professor Carlos Azzoni, whom I have no words to thank for his generosity and trust. He taught me more than economics: he taught me to think with responsibility, write with care, and respect doubt as part of the process. His guidance was a privilege and an inspiration.

To everyone who was with me, my deepest gratitude. If this work holds any value, it is because it carries within it a little bit of each of you.

*“Adversidade? Não, não me venham falar em adversidade.
Diante dela, só há três atitudes possíveis: enfrentar, combater e vencer.”*

Mário Covas

*“Adversity? No, do not speak to me of adversity.
In the face of it, there are only three possible attitudes: face it, fight it, and overcome it.”*

Preface

This work originated from theoretical and practical inquiries that shaped my background in Economics and were consolidated through the analysis of Brazilian labor market dynamics. Throughout the research, it became evident that the mechanisms by which the market adjusts to economic shocks are complex and multifaceted. Therefore, I organized the study into three interconnected essays to address the topic from different perspectives and levels of analysis. To believe that a single isolated approach could capture all nuances would be to oversimplify reality.

The starting point of this work lies in a few central ideas. First, unemployment, although socially painful, tends to be a temporary state in a professional trajectory. What truly matters for the worker's well-being and for the economy is how long this situation lasts and what the costs are to return to the market.

Furthermore, mobility between productive sectors appears as a fundamental adjustment mechanism. When a sector faces difficulties, the reallocation of workers to other sectors can avoid prolonged unemployment. However, this transition is not cost-free: there is a loss of specific human capital, difficulties in adaptation, and informational asymmetries that affect both productivity and the psychological aspect of the worker.

Another important aspect is the dilemma between maintaining employment and preserving wages. In situations of negative shocks, workers often need to accept wage reductions to keep their occupation. This conflict—between remaining employed or maintaining the wage level—is central to labor market search theory and influences decisions made by both workers and employers.

Finally, the ability of an occupation to transit between different sectors, referred to here as portability, is a decisive factor that modulates all these aspects. Occupations with high portability offer greater flexibility and lower vulnerability to sectoral shocks. The measurement and empirical analysis of portability constitute the main contribution of this work.

The structure of the study reflects this multidimensional approach. The first essay presents the theoretical foundations and proposes metrics to quantify portability using RAIS microdata. The second essay investigates how Brazilian regions respond to idiosyncratic shocks, focusing on the 2015-2016 recession and the COVID-19 pandemic. Finally, the third essay examines the differences in wage response to regional unemployment.

This work contributes to integrating three areas of study that are often treated separately. The results suggest that professional retraining policies should prioritize the development of transversal skills that expand the portability of occupations, rather than focusing exclusively on highly specialized training.

I acknowledge, however, two important limitations. The first is that the analysis is restricted to the formal market, which represents about 60% of the Brazilian workforce, leaving out the informal sector, which is significant in the country. The second is the absence of institutional variables, such as labor legislation and union power, which may also influence adjustment mechanisms. These limitations point to promising paths for future research that can deepen the understanding of flexibility and adjustment processes in the Brazilian labor market.

Resumo

Este estudo analisa a dinâmica do mercado de trabalho brasileiro através de três dimensões inter-relacionadas: portabilidade ocupacional, flexibilidade alocativa e flexibilidade salarial. Utilizando microdados da RAIS/MTE (2010-2023), estima-se a portabilidade para cada uma das ocupações presentes nas CBOs. A flexibilidade alocativa apresenta heterogeneidade regional: regiões economicamente diversificadas como o Sudeste ajustam-se mais eficientemente a crises, com São Paulo atingindo índice de 10,91 em 2023, enquanto estados menos diversificados como Maranhão registraram apenas 1,27. Durante a pandemia (2020), o descasamento setorial atingiu picos no Nordeste (0,9607), revelando ineficiências na realocação intersetorial. Quanto à flexibilidade salarial, identificou-se assimetria na resposta ao desemprego regional: ocupações pouco portáteis exibiram elasticidade mais acentuada (-0,0603), indicando que trabalhadores especializados arcam com custos salariais durante crises. Em contraste, ocupações portáteis mostraram menor sensibilidade (-0,0544), beneficiando-se da mobilidade intersetorial (e consequente flexibilidade alocativa). Conclui-se que políticas de requalificação profissional devem priorizar habilidades cognitivas e transversais, enquanto regiões menos diversificadas demandam investimentos em complexidade produtiva para reduzir custos de ajuste.

Palavras-chave: Mercado de trabalho; Portabilidade ocupacional; Flexibilidade salarial; Realocação setorial.

Abstract

This study analyzes the dynamics of the Brazilian labor market through three interrelated dimensions: occupational portability, allocative flexibility, and wage flexibility. Using RAIS/MTE microdata (2010–2023), occupational portability is estimated for each occupation classified under the CBO. Allocative flexibility exhibits regional heterogeneity: economically diversified regions such as the Southeast adjust more efficiently to crises, with São Paulo reaching an index of 10.91 in 2023, while less diversified states like Maranhão recorded only 1.27. During the COVID-19 pandemic (2020), sectoral mismatch peaked in the Northeast (0.9607), revealing inefficiencies in intersectoral reallocation. Regarding wage flexibility, the response to regional unemployment is asymmetric: occupations with low portability displayed higher elasticity (-0.0603), indicating that specialized workers bear greater wage costs during crises. In contrast, portable occupations showed lower sensitivity (-0.0544), benefiting from intersectoral mobility and the resulting allocative flexibility. The findings suggest that professional retraining policies should prioritize cognitive and transversal skills, while less diversified regions require investment in productive complexity to reduce adjustment costs.

Keywords: Labor market; Occupational portability; Wage flexibility; Sectoral reallocation.

Lista de ilustrações

Figura 1 – Skill scores required for some CBO occupations	19
Figura 2 – Average of social, cognitive, and motor skill scores	20
Figura 3 – Shannon entropy by intermediate geographic region	22
Figura 4 – Shannon Entropy (2023)	24
Figura 5 – Sankey Diagram: Occupations (10) that employ the most among the sectors (10) that employ the most	26
Figura 6 – Portability of occupations	28
Figura 7 – Portability of workers	28
Figura 8 – Variation in formal employment	29
Figura 9 – Proportion of portable occupations, by intermediate geographic region	30
Figura 10 – Average portability by macro-region	31
Figura 11 – Average turbulence of intermediate regions, by macro-region.	43
Figura 12 – Average mismatch of intermediate regions, by macro-region.	44
Figura 13 – Average allocative flexibility of intermediate regions, by macro-region.	44
Figura 14 – Allocative flexibility, by state	46
Figura 15 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region); Total	65
Figura 16 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region); Category 1	66
Figura 17 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region); Category 2	67
Figura 18 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region); Category 3	68
Figura 19 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region); Category 4	69
Figura 20 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region); Category 5	70

Lista de tabelas

Tabela 1 – Share of the Top 1% of Occupations in Total Employment Contracts (2010–2023)	18
Tabela 2 – Mean and median of <i>skills</i> in the 31 most frequent occupations	19
Tabela 3 – Portability index for the 10 most frequent occupations (2023)	27
Tabela 4 – Description of independent variables	32
Tabela 5 – Statistical summary of variables by portability group	32
Tabela 6 – PROBIT model results	33
Tabela 7 – Selected indicators for the 4 states with highest and lowest values in 2023 (2015–2023)	45
Tabela 8 – Employees by portability category	46
Tabela 9 – Turbulence, mismatch, and flexibility. National indices	48
Tabela 10 – Descriptive statistics by year	57
Tabela 11 – Percentage distribution of the employed labor force by occupational group (2010–2023).	71
Tabela 12 – Occupations comprising the most numerous 1% of total occupations classified in the CBO, by formal private ties (2010–2023)	72
Tabela 13 – Total occupations by selected sector (2010–2023)	73
Tabela 14 – Shannon Entropy by macro-region (2010–2023)	74
Tabela 15 – Average portability by macro-region (2010–2023)	74
Tabela 16 – Turbulence Index in Intermediate Regions by Macro-region (2010–2023)	75
Tabela 17 – Mismatch Index in Intermediate Regions by Macro-region (2010–2023)	75
Tabela 18 – Allocative Flexibility Index in Intermediate Regions by Macro-region (2010–2023)	76
Tabela 19 – Largest and smallest percentage growth in regional formal employment by sector (2010–2016)	76
Tabela 20 – Largest and smallest percentage growth in regional formal employment by sector (2016–2021)	77
Tabela 21 – Allocative Flexibility Index by State	78
Tabela 22 – Regression Results for the First Period (2010-2016)	79
Tabela 23 – Regression Results for the Second Period (2016–2021)	80

Sumário

	Preface	5
	Lista de ilustrações	9
	Lista de tabelas	10
1	PORTABILITY	13
1.1	Introduction and literature review	13
1.2	Data and overview of the Brazilian formal labor market	17
1.2.1	Occupational distribution of the labor market	17
1.2.2	Sectoral and regional distribution of the labor market	21
1.3	Portability	25
1.3.1	Methodology and theoretical approach	25
1.3.2	Results	27
1.3.3	Determinants of occupational portability	31
1.4	Conclusion	35
2	ALLOCATIVE FLEXIBILITY	36
2.1	Introduction and Theoretical Framework	36
2.2	Methodology	39
2.2.1	Turbulence	39
2.2.2	Mismatch	39
2.2.3	Allocative Flexibility	40
2.2.4	Occupational Portability	40
2.3	Data and Results	42
2.3.1	Intermediate Regions	42
2.3.2	States	45
2.3.3	Results with portability	46
2.3.4	Explaining Flexibility	47
2.4	Conclusion	50
3	WAGE FLEXIBILITY	51
3.1	Introduction, Literature Review, and Theoretical Approach	51
3.2	Methodology	55
3.3	Data and Results	57
3.3.1	Descriptive Statistics	57
3.3.2	Regression Results	57

3.4	Conclusion	59
	REFERÊNCIAS	60
	APÊNDICES	64
	APÊNDICE A – FIGURES	65
	APÊNDICE B – TABLES	71

1 Portability

1.1 Introduction and literature review

The organization of the labor market is one of the central issues in development economics literature, given its role in determining aggregate productivity and mediating socioeconomic inequalities. The occupational structure can be analyzed from different dimensions—sectoral, spatial, and occupational—each revealing specific aspects of the interplay between labor supply and demand. This perspective is essential for understanding the different effects of economic transformations on the labor market in various regional and economic cycle contexts.

From a sectoral perspective, the Brazilian economy has undergone a pronounced process of structural transformation over the last two decades, characterized by the relative loss of importance of the manufacturing industry in the composition of GDP and formal employment. According to [Oreiro e Feijó \(2010\)](#), this process constitutes premature deindustrialization, as it occurs at medium per capita income levels, unlike the trajectory observed in developed economies. Between 2003 and 2014, the share of the manufacturing industry in gross value added fell from 16.9% to 12.0%, with significant losses, especially in the North (-5.9 p.p.), South (-6.0 p.p.), and Northeast (-3.9 p.p.) regions ([\(MACEDO; PORTO, 2023\)](#)).

The advancement of the service sector in Brazil's productive structure is an example of recent labor market transformations. Between 2003 and 2014, the share of services in the national gross value added increased from 65.8% to 71.2%, also reflecting its growing weight in the generation of formal jobs ([\(MACEDO; PORTO, 2023\)](#)). This growth, although significant for labor absorption, was heterogeneous across the sector's segments. A large part of the expansion was concentrated in lower-complexity occupations, such as commerce, accommodation, and food services, which generally require generalist skills and have high turnover and informality rates. In this context, the labor market structure became more polarized, with the coexistence of high-qualification and high-income occupations restricted to specific sectors and a growing mass of jobs in activities with low technical requirements and scarce social protection. However, it is important to show a counterpoint: [Jacinto e Ribeiro \(2015\)](#) identify the service sector as having higher (and growing) productivity than other sectors, such as the manufacturing industry, for example.

From a regional perspective, a process of economic activity deconcentration is also observed. Studies such as [Saboia \(2000\)](#) and [Saboia \(2013\)](#) highlight that, starting in the 1990s, there was an industrial decentralization in Brazil, with industry moving from traditionally industrialized regions, like the Southeast, to other regions, such as

the South, Northeast, and Center-West. This movement was driven by factors like fiscal wars between states, lower wages in less developed regions, proximity to raw material sources, and the development of Mercosur. As a result, states like Paraná, Ceará, and those in the Center-West region showed great dynamism, receiving new industrial companies and presenting strong employment growth ((SABOLA, 2001)). These transformations in the productive structure and its regional redistribution had significant implications for the organization of the labor market, not only in terms of the quantity of jobs generated but also regarding the quality and compatibility between professional profiles and available vacancies. In this context, the debate on persistent imbalances between labor supply and demand gains prominence, even in scenarios of regional economic dynamism.

A recurring question in the labor market debate is how a significant number of job-seeking workers (unemployed) can coexist with companies that have open vacancies. This apparent paradox can be explained by the phenomenon of *mismatch*: the profiles of those offering and those seeking employment do not coincide. A common type of *mismatch* is spatial: it occurs when there is a geographical disconnection between where workers are located and where job opportunities are concentrated, making it difficult for labor supply and demand to meet. Studies like Silva e Monte (2024) measure job accessibility in metropolitan regions and its implications for the labor market, highlighting that better physical access increases market participation, in addition to reducing unemployment, informality, and the incidence of *overeducation*.

Another relevant form of *mismatch* is that of skills: it occurs when there is an incompatibility between the qualifications or competencies of workers and those required by available vacancies. This can manifest as both a lack and an excess of qualification. Research such as that by Silva e Monte (2024) and Marioni (2021) shows that, in Brazil, this type of incompatibility is present, caused by overeducated workers, that is, individuals who are highly qualified for the requirements of the positions they occupy.

Although approaches like spatial and skills *mismatch* help explain part of the imbalances between labor supply and demand, they do not exhaust the possible sources of friction in the labor market. In particular, they fail to capture a fundamental structural dimension: the way occupations are articulated with the productive structure and their ability to adapt to shocks.

In this sense, a still little-explored dimension becomes relevant: the possibility of sectoral mobility without occupational mobility. Many occupations (such as accountants, drivers, or administrative technicians) are performed in multiple sectors of the economy, while others (such as machine operators or aviation professionals) have strong sectoral specificity. For the latter, transitioning between sectors often also requires an occupational change, which imposes additional barriers to worker reallocation.

Understanding the organization of the labor market increasingly involves analyzing

the characteristics of occupations and their interactions with the productive structure. In a scenario marked by technological changes, economic crises, and transformations in the pattern of sectoral specialization, the concept of occupational portability gains importance. This term broadly designates the functional scope and transferability of occupations between different sectoral or spatial contexts. In other words, it is a way of measuring the versatility and potential mobility of workers in the market.

The concept of portability is related to the diversity of sectors in which a given occupation is present. Occupations with high portability are those distributed across multiple sectors of the economy, indicating a set of tasks or skills that are not specific to a restricted productive context. Thus, portability can be understood as an occupation's ability to move between sectors or regions without substantial loss of function or productivity. In this way, it serves as a *proxy* for the resilience and adaptability of the workforce in the face of structural shocks, such as economic crises or technological transformations, since workers in portable occupations tend to find new opportunities more easily, reducing the time of unemployment or underutilization.

This concept is particularly useful for understanding recent dynamics of the Brazilian labor market. During the 2015-2016 recession and the health crisis caused by the Covid-19 pandemic, the country underwent significant sectoral rearrangements. Sectors such as civil construction and the manufacturing industry lost relative importance, while the service sectors (with more generalist occupations) gained ground. Studies like [Rodrigues-Silveira \(2023\)](#) indicate that these transitions not only altered the structure of formal employment but also shifted the demand for different types of skills, especially regarding the social and cognitive requirements of jobs.

International literature offers theoretical and empirical bases for deepening these analyses. [Autor, Levy e Murnane \(2003\)](#), for example, introduce the idea of the *task content* of occupations to describe how technological transformations favor activities that require analytical or interpersonal skills, while replacing those of a more repetitive or motor nature. In this sense, occupations with a high degree of portability would tend to be less vulnerable to automation and more adaptable to productive restructuring. For Brazil, this is a scenario in which job opportunities will be increasingly linked to the worker's ability to perform non-repetitive tasks that involve analytical, decision-making, or interactive content. Routine operation and control tasks or those involving physical skills tend to lose importance as mechanization and automation deepen in the most diverse sectors of activity ([MACIENTE; RAUEN; KUBOTA, 2019](#)).

Although there are approaches that estimate and analyze the flexibility and resilience of the labor market ([BARROS et al., 1997](#)); [STADUTO; SCHIO, 2010](#)), empirical literature that directly investigates the role of occupational portability as a central dimension for the labor market structure is still scarce.

This article, by exploring data from RAIS-MTE between 2010 and 2023, seeks to fill

this gap by proposing empirical metrics to capture this dimension, articulating them with the occupational and sectoral distribution of the Brazilian workforce. Through a longitudinal and regionally disaggregated analysis, it aims to understand how different occupations are distributed.

1.2 Data and overview of the Brazilian formal labor market

The data source is the Annual Report of Social Information from the Ministry of Labor (RAIS-MTE)¹. The analysis is restricted to private sector contracts, in order to focus on the portion of the labor market that allows for a greater degree of adjustment, given the typical rigidity of the public sector. In each year between 2010 and 2023, each formal worker in the region performs their activities in one of the 2719 occupations, in one of the 87 sectors. The sectoral delimitation follows the National Classification of Economic Activities 2.0 (CNAE 2.0)², at the two-digit level. The occupational delimitation follows the CBO — Brazilian Classification of Occupations³, also from the Ministry of Labor. The geographical analysis is done at the level of the five macro-regions as well as the 133 intermediate geographic regions.⁴

1.2.1 Occupational distribution of the labor market

To assess the concentration of the formal labor market by occupations, the distribution of employment contracts among the most representative occupations of each year was analyzed. Specifically, the occupations (CBOs) belonging to the top percentile (top 1%) in number of contracts were identified annually, and the fraction that these contracts represent in the total formal private employment was estimated. This approach allows verifying whether a small portion of occupations accounts for a disproportionately high share of the labor market.

Table 1 presents the absolute number of these occupations and the corresponding share of these contracts in relation to the total, and Table 12 identifies them. It is noted that this small number of occupations accounts for more than 42% of all contracts present in RAIS.

To better qualify these most frequent occupations, especially in terms of the type of knowledge and skills they require, this work uses indicators built from the methodology originally proposed by Maciente (2013), improved by Júnior (2018) and later updated by Júnior, Azzoni e Chagas (2021). The proposal consists of associating each CBO occupation with a set of skill scores extracted from the American O*NET database (*Occupational Information Network*, from the US Department of Labor), which describes the functional requirements of different professions, such as cognitive skills (e.g., analytical thinking), social skills (such as teamwork), and motor skills (such as manual dexterity).

From the cross-referencing between Brazilian occupations and O*NET data, each CBO occupation received numerical scores representing the degree of requirement in

¹ RAIS - Ministry of Labor

² CNAE - Brazilian Institute of Geography and Statistics

³ CBO - Ministry of Labor

⁴ These are located two levels above the municipal delimitation, according to the structure of the IBGE.

Tabela 1 – Share of the Top 1% of Occupations in Total Employment Contracts (2010–2023)

Year	No. of Occupations (Top 1%)	Share of Contracts (%)
2010	25	42.2%
2011	26	42.8%
2012	25	42.5%
2013	26	43.2%
2014	26	43.2%
2015	26	43.4%
2016	26	43.6%
2017	26	43.4%
2018	26	43.6%
2019	26	43.6%
2020	27	44.1%
2021	27	43.9%
2022	27	43.8%
2023	27	43.8%

Source: Author's elaboration with data from RAIS-MTE

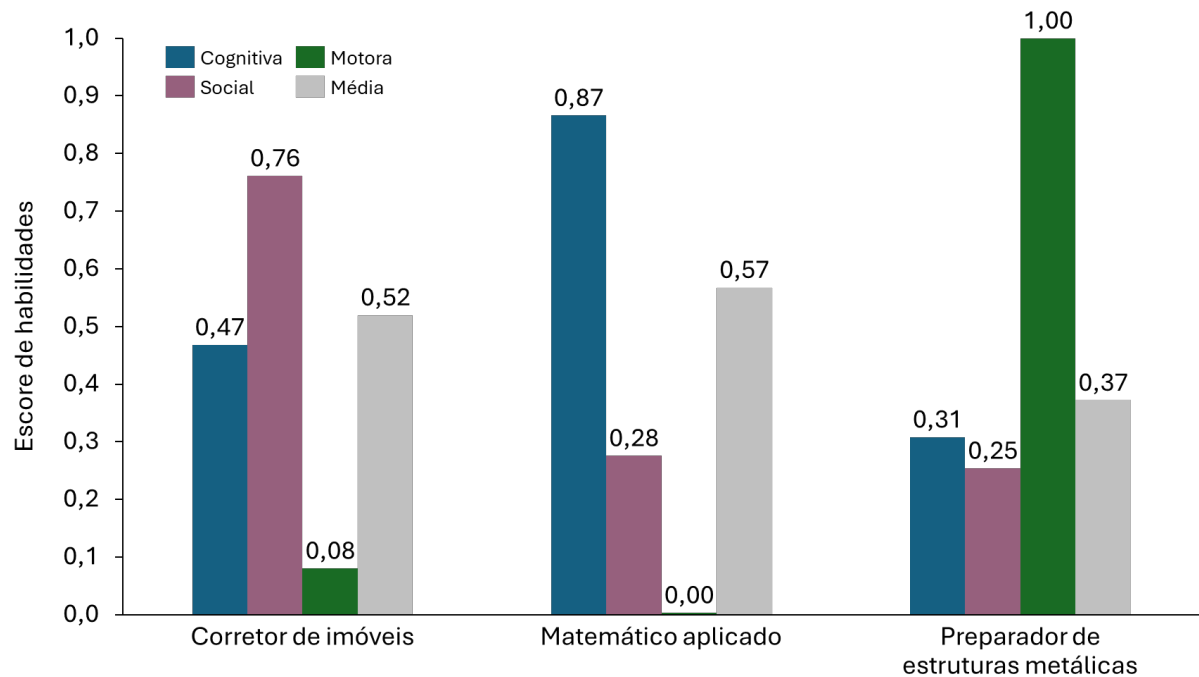
three types of skills: cognitive, social, and motor. These scores range from 0 to 1 and indicate how much a certain skill is required in the exercise of the occupation. In this article, the scores already calculated by Júnior (2018) were used directly, which links these scores to the occupations registered in the RAIS microdata.

Figure 1 illustrates the distribution of skill scores for three representative occupations: real estate agent, applied mathematician, and metal structure assembler. It is observed that these occupations have very distinct requirement profiles: while the applied mathematician has a high demand for cognitive skills (0.87) and very low for motor skills (0.00), the metal structure assembler stands out precisely for its high motor score (1.00), with low cognitive and social requirements. The real estate agent, on the other hand, demands mainly social skills (0.76), although it has intermediate cognitive requirements. These examples help to illustrate how different types of occupations can be distributed along the skill axes.

Based on this structure, Castro (2023) analyzed the evolution of skills in the Brazilian industry between 2003 and 2019. The results show that cognitive and social skills have been gaining more importance, while motor skills tend to lose ground, reflecting technological and productive changes. In addition, the author identified signs of regional convergence: regions that previously employed less sophisticated occupations are gradually approaching the more advanced ones in terms of skill requirements. These findings reinforce the utility of additional indicators as a complement to education in the analysis of the Brazilian formal labor market.

Using this skill score, Table 2 presents the mean and median of the *skills* scores attributed to the 31 distinct occupations that, over the analyzed period, were among

Figura 1 – Skill scores required for some CBO occupations



the most frequent in at least one of the years.⁵

It is noticeable that, in general, the most frequent occupations have low skill requirements, particularly cognitive skills, being less sophisticated in terms of skills. The highest level of requirement occurs in motor skills, a reflection of the operational nature of many of these activities. In contrast, cognitive and social requirements appear with slightly lower values, although relevant in customer service, security, and administrative support functions.

Tabela 2 – Mean and median of *skills* in the 31 most frequent occupations

Skill Dimension	Mean	Median
Cognitive Skill	0.29	0.25
Social Skill	0.30	0.26
Motor Skill	0.39	0.37

Source: Author's elaboration with RAIS data and *skills* score from Júnior (2018)

Figure 2 presents the average *skills* score of occupations. The first graph is the simple average of the *skills* score of the 5% most frequent occupations in the formal labor market in the respective year. The second graph is the average *skills* score of the workers who make up the workforce in these occupations (i.e., a weighted average). The third graph is the average *skills* score of all workers. It is possible to infer from

⁵ The occupation "Store and market attendant" is excluded from the mean and median calculations, as no correspondences were found in the *skills* estimations of (JÚNIOR, 2018).

these graphs that I) occupations and workers are, over the years, becoming more sophisticated in terms of cognitive skills; II) workers are concentrated in occupations that require motor skills, although this is a declining trend; III) the most sophisticated occupations in terms of cognitive skills score are certainly not concentrated much in the top 5%, given the difference in the average cognitive skills score between workers in the top-5% occupations and workers in general.

Figura 2 – Average of social, cognitive, and motor skill scores

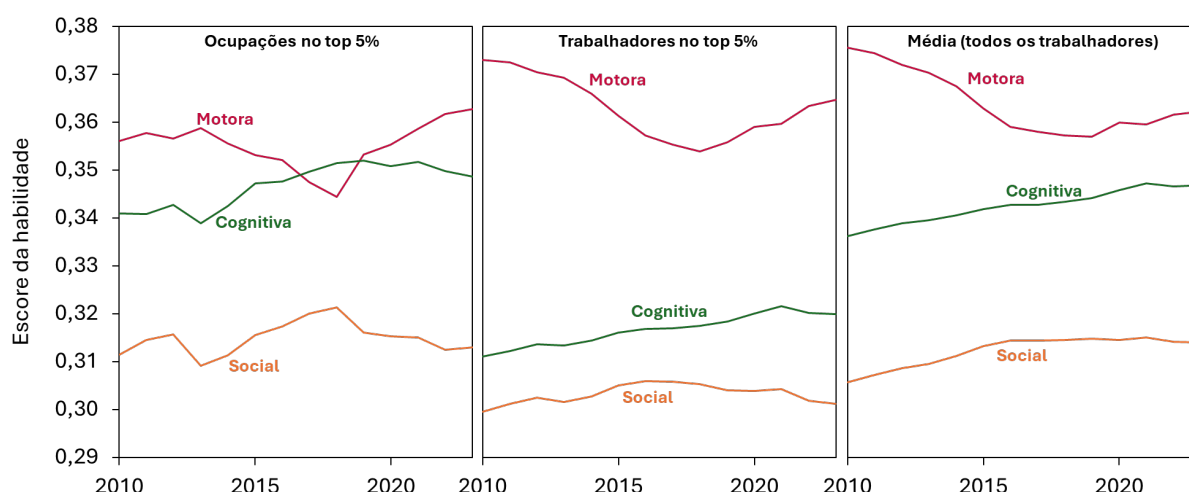


Table 11, in the Appendix, presents the percentage distribution of the Brazilian formal workforce, according to the major CBO occupational groups, between 2010 and 2023. A set of structural transformations is observed, reflecting, simultaneously, sectoral changes, economic shocks, and technological reconfigurations in the labor market.

Firstly, the significant drop in the participation of "Industrial Goods and Services Production Workers"(Group 7) stands out, falling from 26.0% in 2010 to 21.2% in 2023. This trajectory is aligned with the deindustrialization process documented by Morceiro (2018) and with the data on the relative loss of participation of the manufacturing industry in the Brazilian GDP, especially after the 2015–2016 economic crisis.

In contrast, "Service Workers, Retail and Wholesale Salespersons"(Group 5) increased their share from 25.3% to 27.1% in the period. This growth is associated with the relative expansion of the service sector, whose resilience to crises, especially the 2015–2016 recession and the Covid-19 pandemic, is widely recognized in the literature ((ASSUNÇÃO; RIBEIRO; MOURA, 2024)). Occupations in Group 5 are generally concentrated in sectors with lower technical requirements and greater labor flexibility, which contributes to their rapid absorption of labor during periods of economic recovery.

The "Agricultural, Forestry, and Fishery Workers"(Group 6) maintained a relatively

stable participation, fluctuating slightly below 4.4%. Although the agricultural sector has maintained its importance in the GDP, its capacity for formal job creation remained limited, due to continuous increases in productivity and mechanization (([FERREIRA et al., 2006](#))).

1.2.2 Sectoral and regional distribution of the labor market

The analysis of the sectoral diversity of regional employment is essential to understand the economic dynamics and the adaptation potential of different regions in the face of shocks. Studies such as [Giovanini, Pereira e Almeida \(2022\)](#) demonstrate that regions with greater productive diversity tend to have better economic performance, highlighting the relevance of diversification for regional development.

A widely used way to measure diversity is through Shannon entropy, originally developed within information theory ([SHANNON, 1948](#)) and later incorporated into economic studies on the sectoral distribution of firms and productive activities (([JACQUEMIN; BERRY, 1979; FRENKEN; OORT; VERBURG, 2007](#))). Entropy quantifies the degree of uncertainty or unpredictability associated with the distribution of a variable and is particularly useful for assessing the proportional dispersion of workers among different sectors of the economy.

In a regional context, this measure allows capturing the degree of heterogeneity of the productive structure. Consider a region r , composed of $s = 1, \dots, S_r$ sectors with active formal contracts. The relative share of sector s in the total contracts of the region is given by:

$$p_{rs} = \frac{E_{rs}}{E_r} \quad (1.1)$$

where E_{rs} represents the number of formal contracts in sector s of region r , and $E_r = \sum_{s=1}^{S_r} E_{rs}$ is the total of formal contracts⁶ observed in the region. The Shannon entropy associated with the sectoral distribution of employment in region r is then defined as:

$$H_r = - \sum_{s=1}^{S_r} p_{rs} \ln(p_{rs}) \quad (1.2)$$

This index reaches its maximum value when the distribution of workers is perfectly equitable among the sectors, that is, when $p_{rs} = \frac{1}{S_r}$ for all s . In this case, the entropy assumes the value of:

$$H_r = - \sum_{s=1}^{S_r} \frac{1}{S_r} \ln\left(\frac{1}{S_r}\right) = \ln(S_r) \quad (1.3)$$

indicating maximum diversity.

On the other hand, entropy tends to zero when there is complete concentration of employment in a single sector s^* , that is, $p_{rs^*} \rightarrow 1$ and $p_{rs} \rightarrow 0$ for $s \neq s^*$. This follows

⁶ In this article, only formal contracts from the private sector of the labor market are considered.

from the fundamental properties of the natural logarithm:

$$\lim_{x \rightarrow 0^+} x \ln(x) = 0 \quad \text{and} \quad \ln(1) = 0, \quad (1.4)$$

which imply that:

$$H_r \rightarrow - \left(1 \cdot \ln(1) + \sum_{s \neq s^*} \lim_{p_{rs} \rightarrow 0^+} p_{rs} \ln(p_{rs}) \right) = 0 \quad (1.5)$$

Thus, the more concentrated the distribution of workers, the lower the value of H_r , reflecting a lack of diversity or extreme productive specialization.

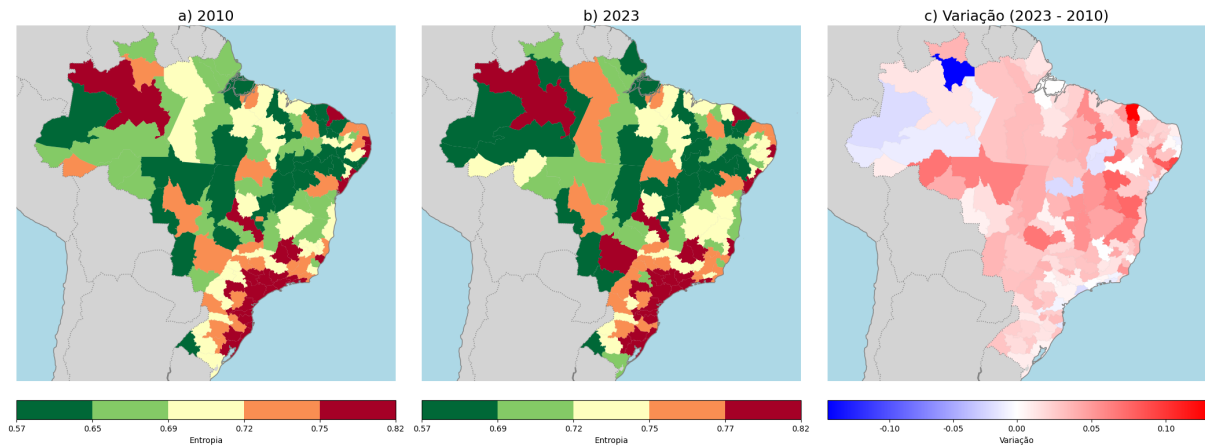
To allow comparisons between regions with different numbers of active sectors, the normalized version of the index is used:

$$H_r^{\text{norm}} = \frac{H_r}{\ln(S_r)} \quad (1.6)$$

so that $H_r^{\text{norm}} \in [0, 1]$. Values close to 0 indicate high concentration of employment in a few sectors, while values close to 1 reflect a more distributed productive structure, with a more balanced sectoral participation.

Therefore, entropy was calculated for each of the 133 intermediate geographic regions, and their values are represented in [Figure 3](#).

Figura 3 – Shannon entropy by intermediate geographic region



Panel (a) represents the Shannon entropy in 2010; panel (b), in 2023; and panel (c), the variation between the years (2023 - 2010). In panels (a) and (b), shades of green indicate less sectoral dispersion (higher employment concentration and less diversification) while shades of red indicate greater dispersion (lower employment concentration and greater sectoral diversity). In panel (c), shades of red indicate an increase in entropy over the period, and shades of blue indicate a reduction.

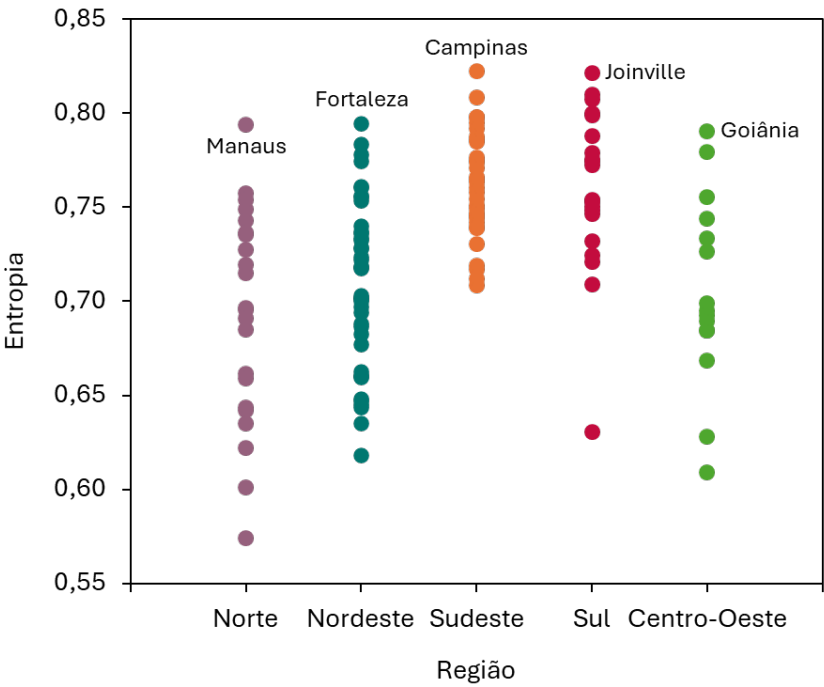
Observing [Figure 3](#), it is possible to identify that, in 2010, the intermediate regions of the Southeast and South already showed high sectoral diversity, a pattern that was

maintained in 2023. However, it is observed that the other regions of the country also expanded their diversity and the distribution of the labor market among sectors, as shown by panel (c) of [Figure 3](#). The evolution of these regions over time reveals a convergence of sectoral diversity, especially in the North and Northeast regions. However, it is noteworthy that regions that include capitals, such as Manaus (AM) and Rio Branco (AC), even when surrounded by areas of low entropy, were already configured as islands of diversity in 2010.

In 114 of the 133 regions analyzed, an increase in entropy was observed, indicating greater dispersion among employment sectors. The most significant increases occurred in regions of the Northeast, such as Sobral (CE) and Maceió (AL), where the "Retail trade" sector became the one with the highest share of formal jobs, a position previously held by the "Preparation of leather and footwear" and "Manufacture of food products" sectors, respectively. On the other hand, some locations showed a drop in entropy, such as Rorainópolis-Caracaraí (RR): the most predominant sector in the region in 2010 was "Retail trade" (29% share of formal employment), surpassed by the "Agriculture, livestock and related services" sector (38% share of formal employment), the same occurred with the region of Gurupi (TO), also in the North Region. Some capital regions, such as Rio de Janeiro (RJ), Salvador (BA), and São Paulo (SP), also had a reduction in their entropy, but in a much more marginal way, and which was not followed by a change in the predominant sector, that of "Retail Trade".

[Figure 4](#) provides a visual inspection of how the South and Southeast regions are the most economically diversified. It can be noted that the South and Southeast regions are more diversified, on average, than the other regions, and the regions with the highest entropy can be identified as large population centers of each region.

Figura 4 – Shannon Entropy (2023)



1.3 Portability

1.3.1 Methodology and theoretical approach

So far, a static analysis of the labor market has been carried out, without considering that, just as sectors are distributed among regions, different occupations are distributed among sectors. The central point is that this distribution is not uniform. [Figure 5](#) shows how the largest occupations are distributed among the largest sectors, combining the occupational and sectoral perspectives. For illustrative purposes, ten very frequent occupations are presented, located on the left side of the figure. On the right side are the sectors in which they are present, with the thickness of the connections indicating the volume of the flow of workers.

Although some occupations show large numbers of workers, they are not uniformly distributed among the sectors. In summary, there are occupations that can be in several sectors, while others have *matches* only with more specific sectors (For example, "Retail salesperson" is the largest occupation in number of workers, but its distribution is basically limited to one sector; while the occupations "Office assistant" and "Administrative assistant" are more uniformly distributed). Therefore, in the face of a contraction in economic activity, it is the type of worker's occupation that influences the speed of reintegration into the labor market, which can make this return faster or slower.

Therefore, the portability of an occupation is defined as its ability to perform the same tasks in different sectors. [Darougheh \(2024\)](#) presents a similar concept, using the term *broadness*.

For the analyses that follow in this work, from the microdata of RAIS between 2010 and 2023, an indicator of occupational-sectoral portability was constructed that follows the methodology proposed by [Darougheh \(2024\)](#). Conceptually, the indicator reflects the sectoral diversity of use of each occupation, measuring how dispersed it is among different productive sectors. For each occupation o , the relative share of employment of that occupation in sector s is defined as

$$p_{os} = \frac{E_{os}}{\sum_s E_{os}} \quad (1.7)$$

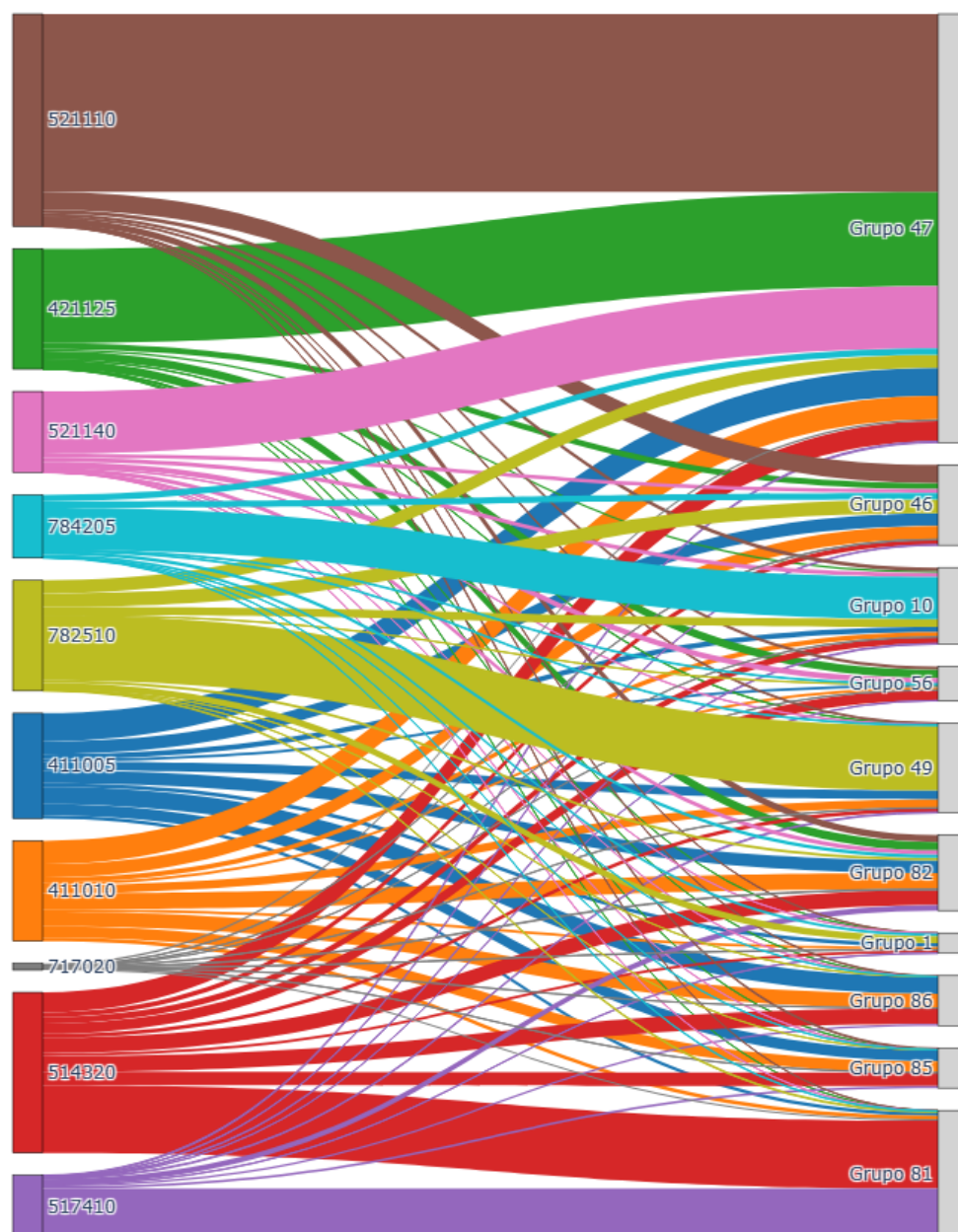
where E_{os} represents the number of jobs of occupation o in sector s . Thus, p_{os} corresponds to the proportion of occupation o in sector s , that is, the fraction of workers of that occupation allocated in that sector.

From this distribution, the portability index of occupation o is defined as

$$m_o = 1 - \sum_s p_{os}^2 \quad (1.8)$$

which corresponds to the complement of the Herfindahl-Hirschman index, traditionally used to measure concentration. In this context, m_o assumes higher values

Figura 5 – Sankey Diagram: Occupations (10) that employ the most among the sectors (10) that employ the most



Note: Occupations (left): 411005 - Office assistant, 411010 - Administrative assistant, 421125 - Cashier, 514320 - Cleaner, 517410 - Building porter, 521110 - Retail salesperson, 521140 - Store and market attendant, 717020 - Construction laborer, 782510 - Truck driver (regional and international routes), 784205 - Production line feeder. **Sectors (right):** 1 - Agriculture, livestock and related services, 10 - Manufacture of food products, 46 - Wholesale trade, except of motor vehicles and motorcycles, 47 - Retail trade, 49 - Land transport, 56 - Food and beverage services, 81 - Services to buildings and landscape activities, 82 - Office administrative, office support and other business support activities, 85 - Education, 86 - Human health activities.

when the workers of occupation o are distributed more evenly among the sectors of the economy, reflecting greater versatility and, therefore, a higher degree of occupational-sectoral portability. The use of the exponent 2 in the sum implies that high values of p_{os} (i.e., high concentrations in a few sectors) have a greater negative weight on the index, penalizing less diversified occupations and accentuating the sensitivity of the index to sectoral concentration.

The index m_o varies between zero and values close to one. It assumes the minimum value of zero when the occupation is entirely concentrated in a single sector, that is, there is a sector s^* such that $p_{os^*} = 1$ and $p_{os} = 0$ for $s \neq s^*$, because in this case $m_o = 1 - 1^2 = 0$, indicating low portability. On the other hand, m_o approaches one when the occupation is uniformly distributed among N sectors, so that $p_{os} \approx \frac{1}{N}$, which implies

$$m_o \approx 1 - \frac{1}{N} \quad (1.9)$$

indicating high portability, as workers with this occupation can move between various sectors with ease.

This indicator was calculated separately for each year of the sample, allowing the observation of the temporal evolution of the portability capacity of occupations.

1.3.2 Results

As a result, each occupation is linked to its respective portability index. [Table 3](#) presents the portability indices for each of the 10 most frequent occupations in the Brazilian formal labor market.

Tabela 3 – Portability index for the 10 most frequent occupations (2023)

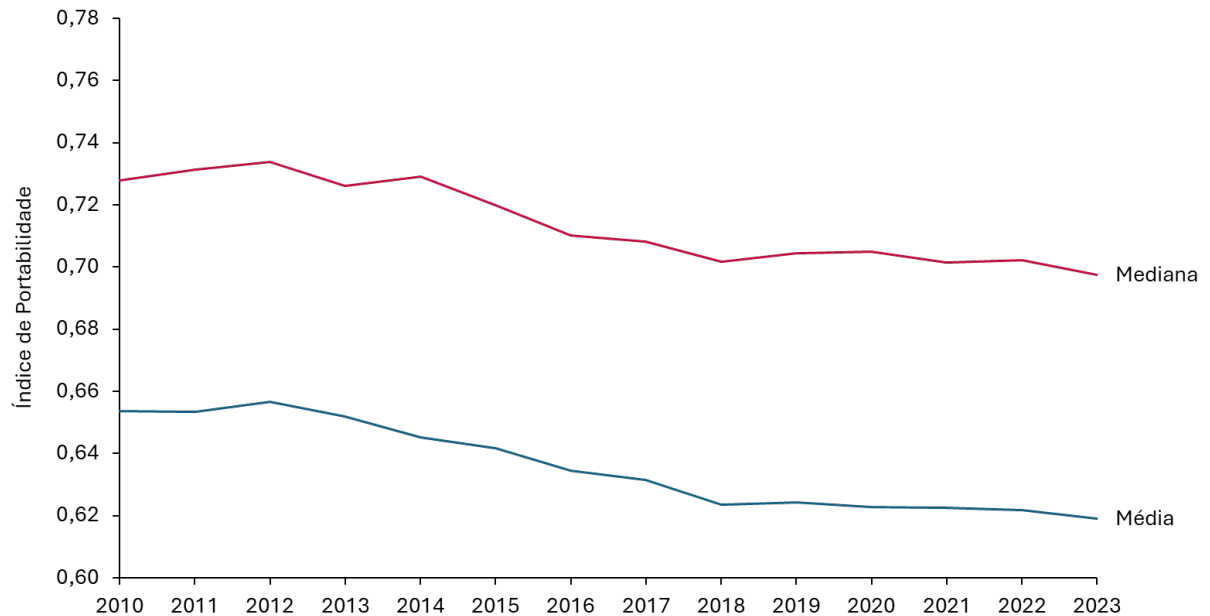
Occupation	Portability Index
Administrative assistant	0.957
Office assistant	0.950
Production line feeder	0.911
Cleaner	0.885
Truck driver	0.779
Construction laborer	0.755
Building porter	0.632
Retail salesperson	0.501
Store and market attendant	0.489
Cashier	0.453

Source: Author's elaboration.

The CBOs present, on average, relative occupational portability. [Figure 6](#) presents the mean and median of the occupations. Two trends are noted: I) the fall of the index

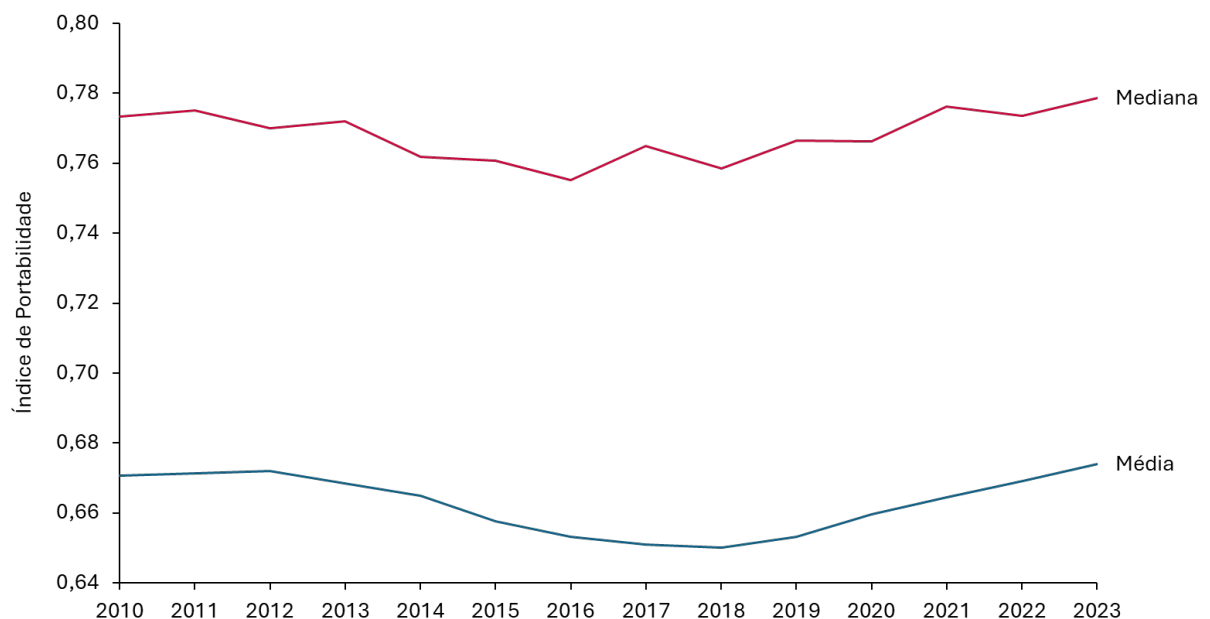
starting in 2012 and ending in 2017 and II) the slight stability of the mean and median from 2018 onwards.

Figura 6 – Portability of occupations



While [Figure 6](#) presents the mean and median of the occupations in the year, [Figure 7](#) presents that of the workers given their occupation. In it, we can perceive that workers are concentrated in more portable occupations and, in the long run, their trend has remained stable.

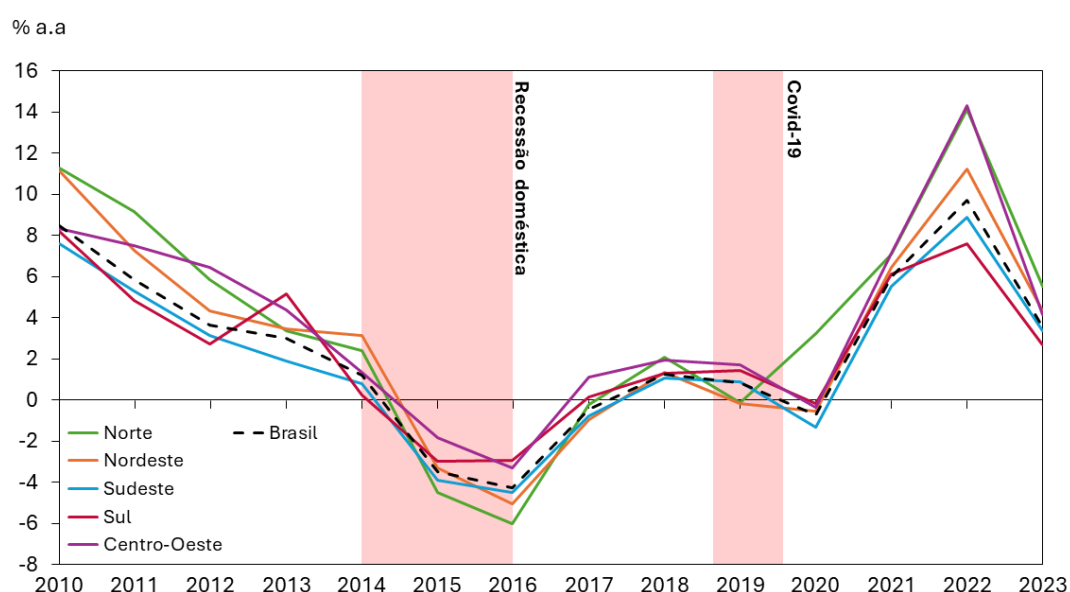
Figura 7 – Portability of workers



A slight reduction in the average portability of occupations is observed during the 2015–2016 recession. This movement suggests a greater concentration of employment in less portable occupations, possibly due to the simultaneous contraction of multiple sectors of the economy. In contexts of aggregate shocks, such as the one that occurred in this period, the capacity for sectoral reallocation of workers is limited, affecting especially those in occupations with greater portability, who, despite their versatility, face additional difficulties in transitioning between sectors in crisis, given that all are contracting.

In addition, the observed drop in the average portability of occupations from 2012 onwards (Figure 6) may be associated with the slowdown in formal employment growth, as illustrated in Figure 8. This scenario reflects a structural change in the Brazilian labor market, where the reduction in the dynamics of formal job creation limits opportunities for sectoral mobility and the entry of new workers.

Figura 8 – Variation in formal employment



The results of Longhi e Taylor (2013) for Great Britain on occupational mobility⁷ of employed and unemployed workers offer a good framework for the portability metric of this article. They show that, in both groups, only about 30% of individuals remain in the same occupation, while more than half make a significant change between major occupational groups. However, the trajectories are very different: employed workers are more likely to progress to higher-category occupations and less likely to regress to lower categories, while the unemployed show precisely the opposite. This suggests that,

⁷ Portability gets its name from the worker's ability to carry their skills across different sectors, and mobility refers to the worker's change of occupation.

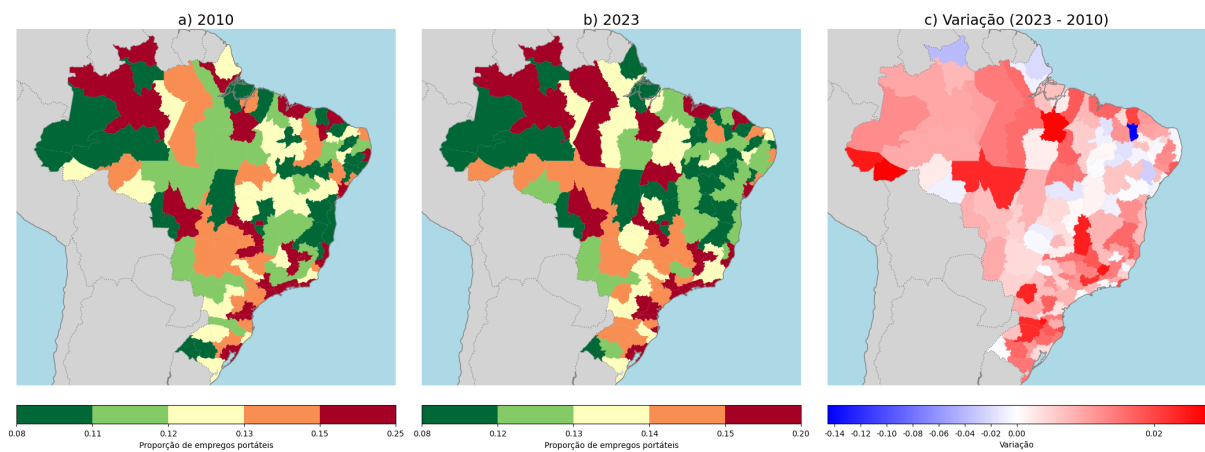
for the unemployed, changing occupations tends to reduce wage prospects, whereas, for the employed, it is generally associated with remuneration gains.

These patterns demonstrate that portability (or *broadness*) goes beyond a simple index of sectoral dispersion: it also anticipates opportunities for wage progression. The more portable an occupation is (i.e., the more distributed among sectors), the greater the probability that an employed worker will obtain wage gains in case of a transition, unlike the unemployed, for whom this change usually represents a setback.

Furthermore, in times of crisis, workers in occupations with a high degree of portability can be re-employed more quickly without even changing occupation, minimizing income and career time losses. Thus, the index m_o not only measures the sectoral versatility of occupations but also functions as a *proxy* for the wage and temporal resilience of workers in adverse contexts ((LONGHI; TAYLOR, 2011a); (LONGHI; TAYLOR, 2011b);(LONGHI; TAYLOR, 2013)).

A highly portable occupation is defined as one that is in the 95th percentile of the portability distribution, i.e., among the 5% of occupations with the highest values of m_o . Figure 9 presents the proportion of jobs occupied by workers in portable occupations, disaggregated by intermediate geographic region. Panel (a) shows the data for the year 2010; panel (b), for the year 2023; and panel (c) shows the variation in the proportion of portable jobs between (2023 - 2010).

Figura 9 – Proportion of portable occupations, by intermediate geographic region

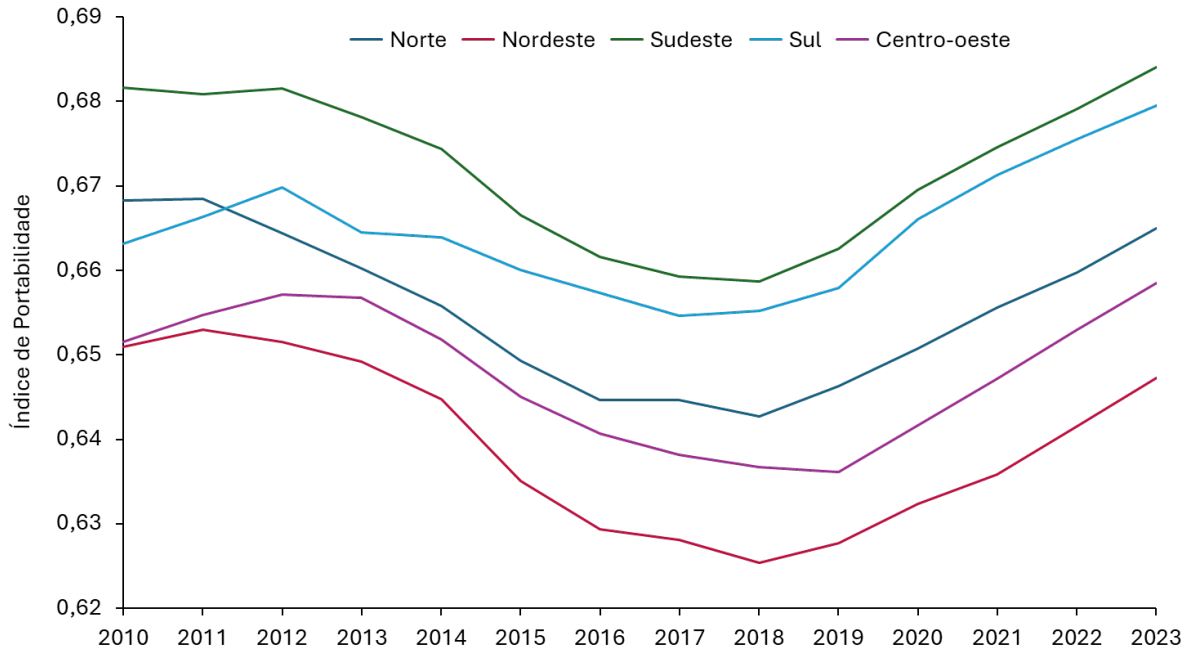


It is noted that Sobral (CE) and Maceió (AL), two regions that increased their diversity, are also among the list of regions that increased their proportion of portable jobs.

In terms of macro-regions (Figure 10), consistent with the previous results, the indices of the Southeast region are the highest, followed by those of the South region, with the Center-West region not far behind the two; but the North and Northeast regions are. A growing trend in the average portability of all regions is observed, especially post-pandemic. However, the trend is clearly more pronounced in the North

and Center-West regions, specialized in mineral resources and agricultural activities, respectively. It is noted that the gap between the average portability of the Southeast and the Northeast has increased considerably in the period. The values can be found in the Appendix, in [Table 15](#).

Figura 10 – Average portability by macro-region



1.3.3 Determinants of occupational portability

To analyze the factors that influence the probability of a worker being in a highly portable occupation, a Probit model was estimated. The dependent variable is a binary variable that takes the value 1 when the worker is in an occupation located in the 95th percentile of the portability distribution (i.e., among the 5% most portable) and 0 otherwise. Therefore, in 2023, an occupation was considered highly portable if its portability index (m_o) was greater than 0.95.

The model formulation is given by:

$$P(\text{Portable}_i = 1) = \Phi(\mathbf{X}_i\boldsymbol{\beta})$$

where $\mathbf{X}_i$ represents the vector of explanatory variables for worker i and $\Phi(\cdot)$ is the cumulative distribution function of the standard normal.

[Table 4](#) presents the variables included in the model, while [Table 5](#) presents their respective descriptive statistics.

As a result, we have the estimated coefficients in [Table 6](#). The estimates obtained reveal patterns consistent with the literature on occupational segmentation and productive structure. The model indicates that individuals with higher remuneration,

Tabela 4 – Description of independent variables

Variable	Description
Age	Worker's age (in log)
Remuneration	Average remuneration in the year (i.e., captures variations), in minimum wages (in log)
Establishment size	Number of employees in the company where the worker is employed
Sex	Dummy: 0 for female, 1 for male
Education level	Dummy: 1 if has complete higher education or more, 0 otherwise
Cognitive score	Score of cognitive skills associated with the occupation
Social score	Score of social skills associated with the occupation
Motor score	Score of motor skills associated with the occupation
Diversification	Entropy index of the region where the worker is located (H_r^{norm})

Source: Author's elaboration with RAIS-MTE documentation.

Tabela 5 – Statistical summary of variables by portability group

Variable	Non-portable	Portable (Top 5%)
Average age	37.072	33.924
Average remuneration (in minimum wages)	4.556	16.144
Average establishment size	5.973	6.098
Proportion with higher education	13.9%	28.1%
Proportion of men	60.8%	42.9%
Average cognitive <i>skills</i> score	0.338	0.439
Average social <i>skills</i> score	0.312	0.347
Average motor <i>skills</i> score	0.383	0.162
Average entropy	0.772	0.775
Number of observations	36,216,286	4,495,804

Source: Author's elaboration with RAIS-MTE data

older age, male, and with complete higher education have a lower probability of being in highly portable occupations, as defined in this work.

These findings indicate that the most portable occupations, on average, require less formal education and are less linked to highly specialized career paths, often concentrated in specific sectors of the economy, hypotheses already outlined in this work. Greater portability manifests, therefore, in occupations with a transversal profile, such as administrative technicians and operational assistants, whose functions have a generalist application in various economic segments. This characteristic is consistent with the idea that support occupations have greater flexibility of sectoral allocation, since their requirements are less specific and widely demanded. According to [Gathmann e Schönberg \(2010\)](#), occupations that share similar sets of tasks facilitate the transfer of human capital, which reduces mobility costs and favors reallocations in different

Tabela 6 – PROBIT model results

	<i>Dependent variable:</i>
	Portable
γ (Education level)	−0.261*** (0.001)
δ (Sex)	−0.075*** (0.001)
α (Age)	−0.308*** (0.001)
β (Remuneration)	−0.002*** (0.001)
θ (Establishment size)	0.003*** (0.0001)
η (Diversification)	0.346*** (0.009)
ψ (Cognitive score)	3.637*** (0.003)
ϕ (Social score)	−3.758*** (0.003)
κ (Motor score)	−3.560*** (0.002)
Constant	0.532*** (0.008)
Observations	39,257,938
Log-Likelihood	−10,333,817.000
Akaike Inf. Crit.	20,667,654.000

Source: Author's elaboration. *Note:* Robust standard errors in parentheses.

Statistical significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

contexts. In a complementary way, [Autor \(2013\)](#) argues that intermediate qualification functions, which combine technical, interpersonal, and adaptive skills, tend to be more resilient to automation and technological changes, precisely because they involve tasks that are difficult to replace or outsource. Both approaches converge on the idea that occupational portability is associated with the presence of broad, less specific, and replicable tasks in different sectors of the economy.

The positive and statistically significant coefficient of the cognitive skills score indicates that more portable occupations require greater cognitive content⁸. This finding partially corroborates findings in which the fastest-growing occupations simultaneously combine cognitive and social skills, giving them high adaptability and transversal applicability ([DEMING, 2017](#)). In contrast, the scores for social and motor skills showed a negative association with portability, suggesting that functions with a greater interpersonal or physical component tend to be concentrated in specific sectors, such

⁸ But note that, as previously mentioned, this score is not something extremely required in these more portable occupations. An alternative to isolate the effect in future work would be to estimate the index with pairs of *skills* scores

as health, education, or civil construction, which restricts their intersectoral mobility⁹.

The regional sectoral entropy variable, which measures the economic diversity of the location, also showed a positive association with portability. More diversified regions tend to offer a more conducive environment for sustaining versatile occupations, favoring adaptability and possibilities for transversal reallocation, as discussed by [Faggio, Salvanes e Reenen \(2010\)](#) and [Neffke et al. \(2011\)](#).

Additionally, the size of the establishment showed a positive relationship with the probability of a worker being in a highly portable occupation. This result may reflect the fact that larger firms, by operating on multiple fronts or demanding a greater variety of support functions that are not related to the core business, more frequently absorb this type of occupation.

These results align with the literature and reinforce the centrality of portability in the configuration of regional and national employment. [Santos et al. \(2023\)](#), when analyzing the effects of the 2015/2016 crisis on formal employment in Minas Gerais, found that workers whose previous occupations and sectors had greater affinity with the local productive structure returned to the labor market more quickly, highlighting the role of compatibility between previous experience and regional demand for skills in professional reintegration.

Finally, [Jara-Figueroa et al. \(2018\)](#) and [Farinha et al. \(2019\)](#) show that specific industry and location knowledge, even more so than occupational knowledge, is fundamental for the success of new firms in regions undergoing economic consolidation. This suggests that, in times of crisis, the reallocation capacity of workers strongly depends on the existence of portable occupations and related sectors present in the territory. Thus, by facilitating transitions between sectors and functions, occupational portability contributes to the resilience of the labor market, mitigating the adverse impacts of economic shocks, given that workers have specific industry knowledge of various industries.

⁹ In this case, sales-related occupations may have contaminated the analysis in some way. These professionals have a very high content of social skills but end up concentrating in a few sectors. Furthermore, they are the most present in the workforce.

1.4 Conclusion

This study explored occupational portability in the Brazilian formal labor market, highlighting its relationship with structural transformations, skill demands, and regional dynamics between 2010 and 2023. Using data from RAIS-MTE, indicators were constructed to capture the ability of occupations to be distributed across multiple sectors, revealing distinct patterns of adaptability. Occupations such as administrative assistants and office assistants, with high portability, were associated with cognitive skills and a transversal presence in various sectors, while functions like retail salespersons showed strong dependence on specific productive contexts.

The regional analysis showed that sectoral diversification, measured by Shannon entropy, is positively related to occupational portability, with emphasis on the Southeast and South, historically more dynamic regions. However, a reduction in the diversity gap between more diversified regions and less diversified areas, such as the North and Northeast, was observed, driven by the expansion of sectors like commerce and services.

The determinants of portability, analyzed via a Probit model, indicated that cognitive skills, regional sectoral diversification, and working in larger companies increase the probability of a worker being in portable occupations. In contrast, motor skills, high remuneration, and higher education are related to less adaptable occupations, reflecting a segmentation between generalist and specialized functions. These results echo international findings, which highlight the growing valuation of analytical and interpersonal skills over routine tasks, in line with technological transformations and productive restructuring.

As a limitation, the study was restricted to the formal market, underestimating informality, a prominent feature of the Brazilian context. Future research could expand the analysis to include informal workers, in addition to exploring the interaction between portability, occupational mobility, and wage inequality. Another relevant front would be to investigate how automation affects occupations with different skill profiles, especially in sectors in relative decline, such as the manufacturing industry.

In summary, occupational portability reveals itself as a fundamental dimension for understanding the dynamics of the labor market in economies undergoing structural transformation. Its ability to mediate economic shocks, combined with its relationship with cognitive skills and productive diversification, positions it as an essential metric for analyses of resilience and adaptability in the labor market. This work thus contributes to a research agenda that integrates occupational characteristics and regional heterogeneity.

2 Allocative Flexibility

2.1 Introduction and Theoretical Framework

The functioning of the labor market is subject to constant reconfigurations. Technological changes, macroeconomic fluctuations, demographic transformations, international trade restructuring, public policies, and unexpected events like pandemics continuously alter the level and composition of labor demand. These shocks provoke shifts in the productive structure and pose challenges to the efficient reallocation of workers. Understanding how regional markets respond to such disturbances is essential to evaluate the effectiveness of adjustment mechanisms, as well as the magnitude of social costs resulting from the transition.

In the theoretical field, economic shocks are defined as unexpected changes in production and demand conditions that affect the optimal allocation of resources. When these shocks directly impact labor demand, whether in terms of total volume or sectoral and spatial distribution, responses promoting the recomposition of equilibrium in an efficient and socially acceptable manner become necessary. Two main mechanisms are traditionally recognized in the literature as adjustment instruments in the face of such imbalances: wage flexibility and allocative flexibility.

Wage flexibility refers to the ability of real wages to adjust to changes in market conditions. In an ideal competitive environment, reductions in labor demand should lead to wage drops, while increases in productivity or demand would result in wage increases. This adjustment via prices would tend to reestablish the equilibrium between supply and demand. However, empirical experience demonstrates that wages rarely respond fully and immediately. Institutional factors, such as contract rigidity, union activity, labor legislation, the presence of minimum wages, and social norms determining acceptable remuneration levels, contribute to wages exhibiting nominal and real rigidity, especially in the short term. This rigidity implies that adverse shocks can generate persistent unemployment and underutilization of the labor force, as the adjustment via price becomes limited.

The literature developed since the 1980s emphasized that labor market imperfections are not transient failures but structural characteristics of real market functioning. Models like that of [Shapiro e Stiglitz \(1984\)](#) and subsequent developments by [Pissarides \(2000\)](#) highlight the presence of frictions in the matching process between workers and vacancies, the importance of asymmetric information, and mobility costs as central elements in explaining equilibrium unemployment. In these models, even in situations of compatible supply and demand, the time required for vacancies to be filled and workers to find suitable positions generates persistent misalignments.

In contexts where wages do not adjust fully, the other avenue for recomposing equilibrium gains prominence: allocative flexibility. This concept concerns the economic system's capacity to reallocate workers from declining sectors or regions to expanding areas. This form of adjustment depends on labor force mobility, qualification portability, the quality of transport networks, and the effectiveness of employment intermediation systems. In situations where shocks affect sectors unequally, as in the case of specific technological transformations or changes in consumption patterns, the adjustment requires workers to transit between different occupations, sectors, or territories. In these situations, rigidity in labor allocation translates into higher unemployment levels and lower economic efficiency.

A fundamental point for understanding labor market responses to shocks concerns the heterogeneity of the productive structure and, in particular, the elasticity of labor demand in various economic sectors. As emphasized by [Barros et al. \(1997\)](#), even in the face of an aggregate shock, sectors may react differently if they present distinct labor demand elasticities. Sectors with higher elasticity tend to adjust their employment level quickly in response to changes in wages or demand for goods and services, while sectors with lower elasticity may maintain their employment level relatively stable. This heterogeneity implies that aggregate shocks can manifest empirically as sectoral shocks, generating dispersion in employment dynamics between sectors. Thus, the simple observation of divergence between employment growth rates is not sufficient (exclusively) to infer the nature of the shocks, necessitating consideration of the microeconomic characteristics of labor demand as well.

This distinction was originally problematized by [Abraham e Katz \(1986\)](#), who criticized the approach of [Lilien \(1982\)](#), which had proposed that the sectoral variance of employment growth rates could be used as an indicator of the intensity of asymmetric shocks. According to [Lilien \(1982\)](#), aggregate unemployment could be partially explained by the magnitude of this variance, given that greater discrepancies between sectors would indicate difficulties in labor force reallocation. The proposed metric is given by:

$$\sigma_t^2 = \sum_s \omega_{s,t} (g_{s,t} - \bar{g}_t)^2, \quad (2.1)$$

where $g_{s,t}$ represents the employment growth rate in sector s in period t , $\bar{g}_t$ is the weighted average rate, and $\omega_{s,t}$ is the relative weight of each sector in total employment. This variance captures the dispersion of growth dynamics between sectors and, therefore, the potential intensity of idiosyncratic shocks.

The criticisms of [Abraham e Katz \(1986\)](#) highlight that if sectors respond unequally to the same common shock, the variance will also increase without this signifying the occurrence of a true asymmetric shock. In other words, heterogeneity in labor demand elasticity can generate dispersion patterns that lead to sectoral shocks. This point

reinforces the need for an analysis considering not only the observed variance but also the microeconomic fundamentals of the productive structure.

Advancing this perspective, [Barros et al. \(1997\)](#) proposed an alternative metric to estimate allocative flexibility based on the ratio between market turbulence and the degree of mismatch between labor supply and demand. This measure is expressed by:

$$F_{jt} = \frac{T_{jt}}{D_{jt}}, \quad (2.2)$$

where T_{jt} represents the intensity of shocks affecting the occupational structure of the local market j in period t , while D_{jt} measures the misalignment between the worker supply structure and the prevailing sectoral demand. High values of F_{jt} indicate that the market managed to absorb shocks relatively efficiently, promoting reallocations compatible with new productive requirements. Conversely, reduced values signal structural rigidity, excessive specialization, low mobility, or deficiencies in information about available employment opportunities.

Based on this conceptual framework, the present study investigates how the Brazilian labor market responded to two periods marked by intense and asymmetric shocks: the economic and institutional crisis between 2010 and 2016, and the COVID-19 pandemic between 2016 and 2021. Using RAIS microdata, we intend to evaluate the intensity of shocks faced by regional markets, the magnitude of sectoral mismatch, and the relative adjustment capacity expressed by levels of allocative flexibility.

2.2 Methodology

The concept of allocative flexibility has been discussed in the literature as a reflection of the labor market's adaptive capacity in the face of structural and conjunctural shocks. Pioneering studies like that of [Layard et al. \(2005\)](#) explored this theme in European countries, associating adjustment difficulties with rigidity in worker reallocation. In Brazil, the works of [Barros et al. \(1997\)](#) and [Staduto e Schio \(2010\)](#) were fundamental in applying metrics to measure the regional and sectoral response of the labor market to economic shocks. However, these analyses generally concentrated on metropolitan regions and limited periods¹

This study broadens this scope by investigating regional and sectoral variations based on a consolidated methodology, covering a wider spectrum of regions and economic sectors. The focus lies on two periods marked by significant shocks: the economic crisis between 2010 and 2016 and the pandemic between 2016 and 2021. To measure allocative flexibility, three main indicators are used: turbulence, mismatch, and flexibility, as outlined in [Barros et al. \(1997\)](#). Additionally, we incorporate an occupational-sectoral portability index, as described at the end of this section.

2.2.1 Turbulence

The turbulence index measures the intensity of the shock suffered by the regional productive structure, reflected in the alteration of the sectoral composition of employment. It is defined, for region j in period t , by:

$$T_{jt} = \frac{1}{2} \sum_{s=1}^S |\alpha_{s,t} - \alpha_{s,t-1}| \quad (2.3)$$

where $\alpha_{s,t}$ represents the share of sector s in total employment of region j in year t . High values of T_{jt} indicate large structural changes, suggesting the presence of relevant shocks.²

2.2.2 Mismatch

The mismatch index assesses the misalignment between employment variation in each sector and the region's average variation. Its formula is given by:

$$D_{jt} = \sum_{s=1}^S (\Delta E_{sjt} - \Delta E_{jt})^2 \cdot \alpha_{s,t} \quad (2.4)$$

¹ The study by [Barros et al. \(1997\)](#) shows that Brazil has greater allocative flexibility than developed countries.

² Although the data are annual, it is recognized that productive adjustments occur dynamically and continuously, potentially extending over several periods.

where ΔE_{sjt} represents the employment growth rate in sector s of region j between periods t and $t - 1$, and ΔE_{jt-1} is the region's growth rate. The index is weighted by the relative importance of sectors, reflecting whether the adjustment to the new productive structure occurred in a coordinated manner. High mismatches indicate that sectors reacted disjointedly, hindering efficient labor force reallocation³.

2.2.3 Allocative Flexibility

Allocative flexibility expresses the efficiency with which the labor market responds to idiosyncratic shocks. It is the ratio between the turbulence index and the mismatch index:

$$F_{jt} = \frac{T_{jt}}{D_{jt}} \quad (2.5)$$

High values of F_{jt} suggest that, even in the face of intense shocks (high T_{jt}), the regional market managed to absorb them relatively successfully, without generating large mismatches (low D_{jt}). Conversely, low values indicate rigidity in the sectoral reallocation of labor, even if the productive structure was widely affected.

2.2.4 Occupational Portability

To deepen the analysis of allocative flexibility, an occupational-sectoral portability indicator is estimated, based on the methodology of [Darougheh \(2024\)](#), addressed more fully in [Chapter ??](#). This index aims to quantify the degree of dispersion with which each occupation is utilized across different productive sectors, reflecting its potential for intersectoral mobility.

For each occupation o , the relative share of the occupation in sector s is defined as:

$$p_{os} = \frac{E_{os}}{\sum_s E_{os}} \quad (2.6)$$

where E_{os} is the number of employment relationships of occupation o in sector s . Thus, p_{os} represents the fraction of workers in the occupation allocated to sector s .

Based on this distribution, the portability (or broadness) index of occupation o is calculated by:

$$m_o = 1 - \sum_s p_{os}^2 \quad (2.7)$$

³ Note that this is an alteration regarding the works of [Staduto e Schio \(2010\)](#) and [Barros et al. \(1997\)](#). This alteration is necessary given that the literature generally uses PNAD data, which contains the unemployment variable. In this study, as will be shown below, RAIS data were used, containing only formal employment stock. The alternative, then, was to use the regional aggregate of regional employment growth differentials

This index is analogous to the inverted Herfindahl-Hirschman index: values close to 1 indicate an occupation widely distributed among various sectors (high portability), while values close to 0 reflect strong concentration in few sectors (low portability).

This index will be important in this analysis to examine whether the labor force composition (regarding portability) assists in regional adjustment in any way.

2.3 Data and Results

The data used in this chapter are from the Annual Social Information Report (RAIS) of the Ministry of Labor and follow the same methodology of use and matching with other databases as the data used in [Section 1.2](#).⁴

An important point to analyze is that, as previously mentioned, for allocative flexibility to be possible, it is necessary for sectors in expansion and contraction to exist simultaneously in the labor market. Among the analyzed periods, a relevant reconfiguration in the sectoral dynamics of regional formal employment is observed. As shown in [Table 19](#), between 2010 and 2016, growth stood out in sectors such as veterinary activities (+112.78%) and information technology services (+35.41%). In the period from 2016 to 2021 ([Table 20](#)), the largest percentage increases occurred again in veterinary activities (+131.76%) and in associative organizations (+75.44%), indicating a possible expansion of specialized services and the third sector. In contrast, domestic services showed a sharp drop in both periods (-37.98% and -61.99%, respectively), reflecting structural changes and possibly the effects of the crisis and labor formalization in this segment.

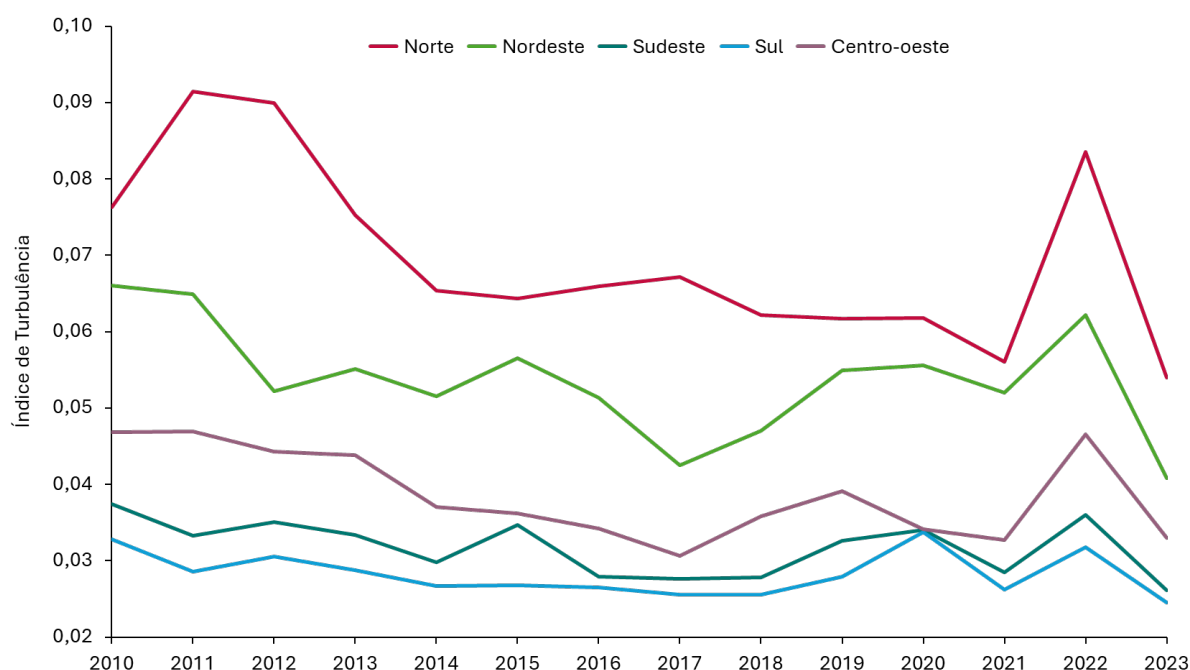
2.3.1 Intermediate Regions

The turbulence index ([Figure 11](#)), measuring the intensity of sectoral changes in employment structure, shows differentiated behavior among regions. The North, for example, presents high levels at the beginning of the series (0.0763 in 2010), a peak in 2011 (0.0914), and a downward trend until 2021, with a new high in 2022 (0.0836), likely related to post-pandemic restructuring. In the Northeast, turbulence remains relatively stable, with slight oscillations, reflecting a pattern of lower sectoral dynamism. The Southeast and South present the lowest turbulence levels, consistent with more consolidated and diversified labor markets. The Center-West shows a slight drop until 2017, followed by a slight rebound in 2022, which may reflect the dynamic role of agro-industry and the service sector in the region.

The mismatch index ([Figure 12](#)), measuring the misalignment between sectoral growth and the regional average, reveals more volatile patterns. The North presents extremely high values at the beginning of the series (0.687 in 2010), falling significantly until 2015 and rising again in 2020 (0.4088), the year in which pandemic shocks deeply disjointed coordination between sectors. The same phenomenon is observed in the Northeast, which presents an abrupt peak in 2013 (0.9128) and another in 2020 (0.9607), indicating strong sectoral mismatch in moments of crisis. The Southeast and South, again, present lower and more stable levels, with values generally below 0.05, indicating

⁴ Throughout the work, to spare the reader from repetition, some methodologies and data uses already explained elsewhere in the work are omitted.

Figura 11 – Average turbulence of intermediate regions, by macro-region.



greater sectoral synchrony. The Center-West shows important fluctuations, notably in 2011 (0.2565) and 2022 (0.0928), suggesting that despite being lower than in the North and Northeast, there are relevant episodes of mismatch⁵.

The allocative flexibility index (Figure 13), which measures the labor market's capacity to adjust to shocks through sectoral reallocations⁶, presents growth trajectories in all regions throughout the analyzed period. The Southeast and South consistently exhibit the highest values, notably the South in 2014 (3.6521) and the Southeast in 2023 (3.4920), indicating more adaptable markets. This pattern is consistent with more sophisticated labor markets and greater factor mobility. In contrast, the North and Northeast, although also showing increased flexibility over time, start from much lower levels and with greater instability, such as peaks in 2016-2017 and 2023. The Center-West shows acceleration especially between 2014 and 2016, with the index jumping from 1.6301 to 2.6079, reflecting probable effects of productive reconfiguration in agribusiness during the crisis.

⁵ This "late" adjustment can be explained, especially, because in 2022 flexibilization measures were broader as well as government aid programs for wage payment were ended

⁶ This is consistent with the discussion on the evolution of sectoral diversification (measured with the Shannon Index) in Chapter ??.

Figura 12 – Average mismatch of intermediate regions, by macro-region.

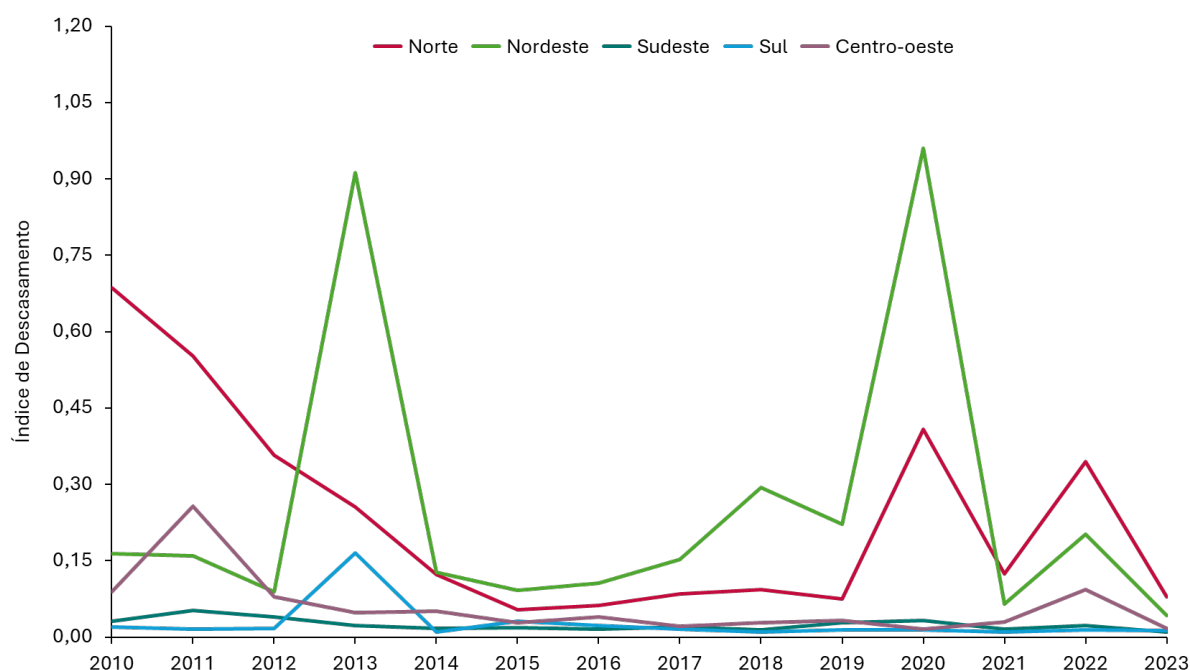
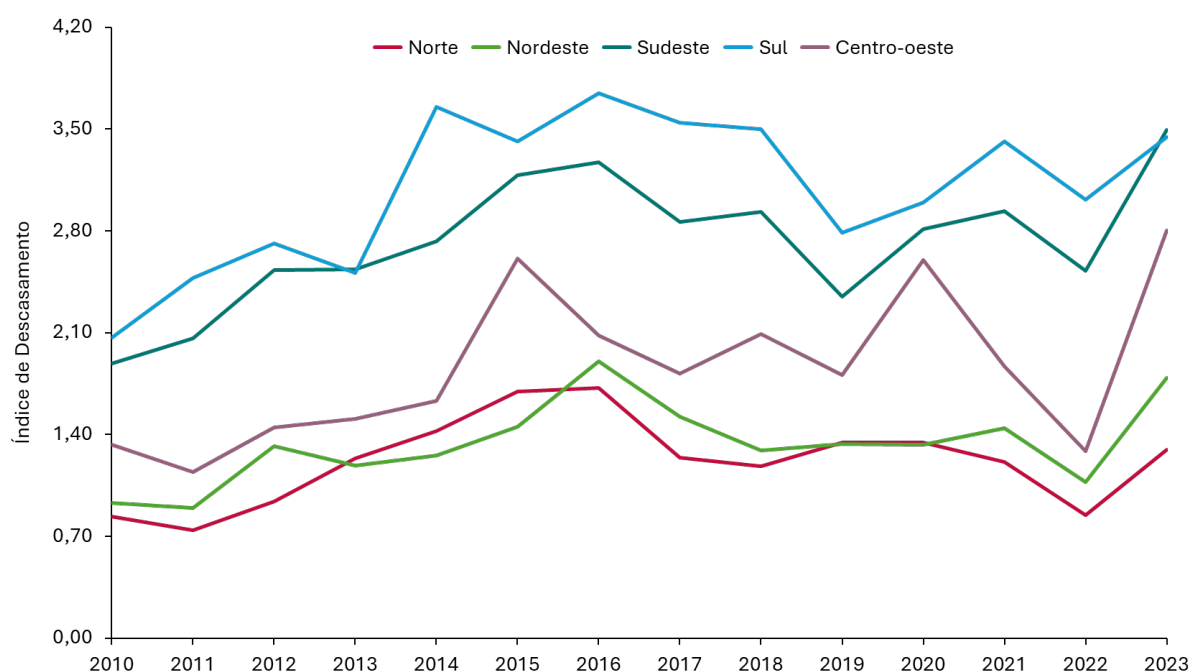


Figura 13 – Average allocative flexibility of intermediate regions, by macro-region.



Generally, the data indicate that economic shocks (such as the 2014-2016 crisis and, especially, the 2020 pandemic) resulted in increased mismatch and turbulence in regions with lower economic diversification and lower flexibility, such as the North and Northeast. On the other hand, regions with more structured labor markets, such

as the South and Southeast, showed greater resilience, with lower volatility and greater adjustment capacity, as evidenced by high levels of allocative flexibility. This pattern reinforces the hypothesis that the degree of complexity and integration of regional markets directly influences their capacity to absorb and react to shocks.

Complete data can be found in [Chapter B \(Table 16; Table 17; Table 18\)](#)

2.3.2 States

Although the results estimated for each intermediate region are interesting, it is important to consider that they omit a relevant labor market variable: migration. Workers may be willing to seek alternatives outside their region to find an employment opportunity. To reduce this migration bias, indices were estimated for each state.

[Table 7](#) presents allocative flexibility data by state for the selected years. Complete data tables are located in [Chapter B](#).

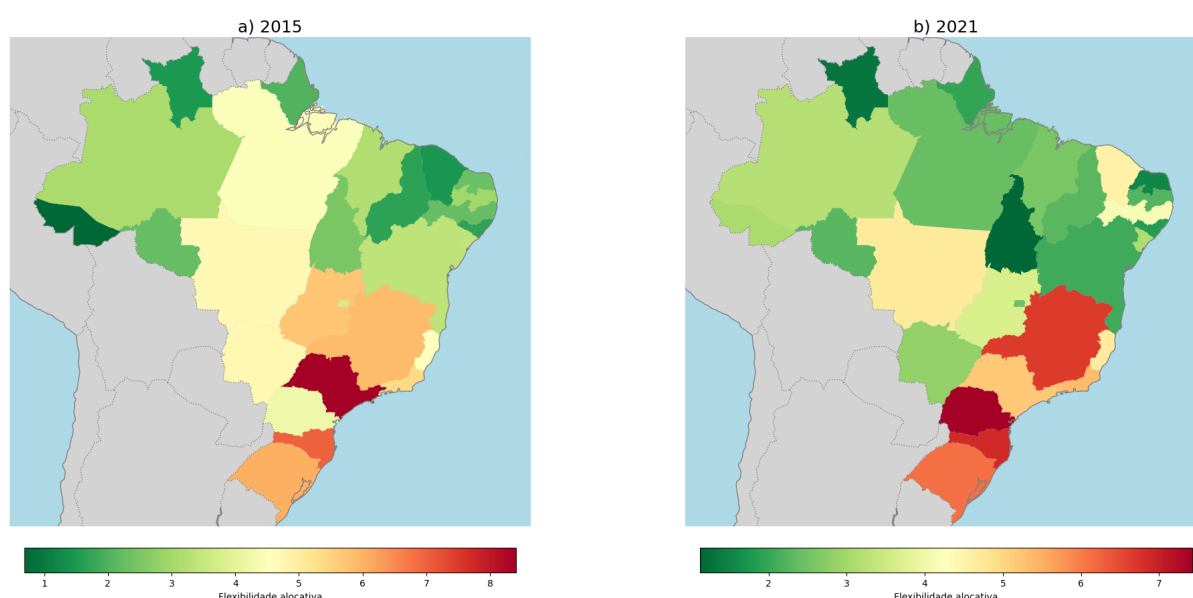
Tabela 7 – Selected indicators for the 4 states with highest and lowest values in 2023 (2015–2023)

UF	2015	2016	2017	2018	2019	2020	2021	2022	2023
São Paulo	8.23	8.41	9.32	7.67	7.68	5.32	3.45	5.18	10.91
Minas Gerais	5.58	5.89	7.02	6.62	8.66	6.25	4.69	6.66	8.49
Espírito Santo	2.05	4.58	4.80	5.06	3.49	3.37	3.13	4.69	6.84
Rio Grande do Sul	1.19	6.08	6.40	4.97	6.49	4.08	4.77	6.12	6.83
Maranhão	2.86	3.16	0.71	3.85	1.26	1.55	1.24	2.59	1.27
Roraima	1.63	1.50	0.94	1.08	1.34	1.86	0.73	1.32	1.29
Alagoas	1.37	1.66	3.53	1.38	0.79	2.30	3.35	1.85	1.41
Amapá	1.00	2.04	1.83	1.46	0.48	0.52	1.37	1.98	1.52

The trajectory of Brazilian states between 2015 and 2023 reveals a dynamic of polarization regarding structural adjustment capacity and labor market flexibility. While states like São Paulo, Santa Catarina, and Paraná started from high levels, they managed to further intensify their allocative flexibility, reaching high values, such as 10.91 for São Paulo in 2023 or 7.42 for Santa Catarina in 2021. These states demonstrated not only resilience but also continuous adaptation capacity facing shocks like the 2015-16 crisis and the 2020 pandemic. On the other hand, a significant set of states with lower dynamism, such as Amapá, Roraima, Alagoas, and Maranhão, remained practically stagnant, oscillating around low values throughout the series. In 2023, for example, Maranhão registered only 0.18, the lowest value among all federative units. The contrast between trajectories indicates that, instead of regional convergence, the period was marked by accentuated divergence: states that were already flexible became even more efficient in reallocating their labor force, while others were left behind. This pattern is indicative of persistent structural barriers limiting mobility and sectoral diversification in regions less integrated into national production chains.

To reinforce this, it is possible to observe in [Figure 14](#) that there is a clear division between regions with more flexible states (Southeast and South) and less flexible ones (Northeast and North).

Figura 14 – Allocative flexibility, by state



2.3.3 Results with portability

With the estimated portability index m_o , workers were divided into five groups, following the quantiles of the portability index distribution⁷. Administrative assistants, accountants, commercial managers, cleaners, and security guards are among the most portable occupations (close to 1). On the other hand, dentist-epidemiologists, speech therapists specialized in orofacial motricity, and trapeze artists are among the least portable occupations (portability index closer to zero). In any case, the relationship between education and portability is complex. Therefore, the portability index presented in this article brings a new viewpoint to labor market analysis. [Table 8](#) presents the number of employees by portability quintile. 1 are the most portable employees, while group 5 is the group of least portable employees.

Tabela 8 – Employees by portability category

Year	General	1	2	3	4	5
2010	34,303,714	28,629,217	3,859,467	1,146,742	468,208	140,873
2016	36,239,523	30,528,504	3,822,796	1,257,716	471,606	146,678
2021	38,774,380	32,906,352	3,925,208	1,319,024	481,148	131,772

⁷ That is, divided into 5 parts with the amplitude from the smallest m_o to the largest m_o , and not relative to the labor force distribution.

Unlike the previous subsection, the analysis concentrated on two specific periods: 2010–2016 and 2016–2021. The Brazilian economy experienced a sequence of eight years of positive growth rates, driven by a favorable international situation, with strong demand and high commodity prices. To support this external demand-driven stimulus, government cash transfer programs for low-income families boosted domestic demand. This favorable period ended around 2010. Thus, the first interval comprises the subsequent sequence of decreasing rates, which involved political turbulence and ended with the president's impeachment in 2016. The following administration managed to correct some of the distortions introduced by the previous government⁸ and the economy began to recover when the COVID-19 shock hit. The recovery from the pandemic was almost immediate⁹.

Detailed regional results for the 133 regions are presented in maps in [Chapter A](#) ([Figure 15](#); [Figure 16](#); [Figure 17](#); [Figure 18](#); [Figure 19](#); [Figure 20](#)).

[Figure 9](#) presents results for all workers and for the five portability levels. Turbulence rates, measuring a region's sensitivity to shock, were higher in the 2010–2016 period across all categories, except the least portable one. The 2015–2016 economic crisis was structural in nature, reducing demand and production in many sectors, leading to higher turbulence. Conversely, the COVID-19 shock reduced output more strongly in the short and medium term, particularly in face-to-face activities. Some occupations, such as lawyers, financial market professionals, and others, continued their activities remotely. As a remedy for the chaotic situation, the government injected resources into the economy to help small businesses and the most vulnerable people.

Without considering portability levels, turbulence was higher in the first crisis, both in terms of mean and median. The same pattern holds for the first four portability layers. Interestingly, the group of occupations with the highest portability reversed the situation, as turbulence was higher in the second period.

2.3.4 Explaining Flexibility

Analyzing how the three indices behave at different portability levels, a loss of locational pattern is observed as the portability level decreases. To investigate the determinants of flexibility in each period, a regression was estimated with some variables of interest. The regression was estimated for each of the 5 portability categories and can be represented by the equation below:

$$\log(F_{cjt}) = \beta_0 + \beta_1 * \log(PPC_{cjt}) + \beta_2 * \log(HH_{cjt}) + \beta_3 * \log(N_{cjt}) + \epsilon \quad (2.8)$$

⁸ Especially a major [labor reform](#), reducing some costs and introducing more flexibility for informal workers.

⁹ This description can be better absorbed by observing [Figure 8](#), in [Chapter ??](#).

Tabela 9 – Turbulence, mismatch, and flexibility. National indices

Portability		Turbulence		Mismatch		Flexibility	
		2010-16	2016-21	2010-16	2016-21	2010-16	2016-21
General	Mean	0.171	0.128	0.766	1.181	1.139	1.337
	Median	0.147	0.115	0.072	0.046	1.068	1.075
1	Mean	0.161	0.126	0.610	0.460	1.069	1.298
	Median	0.134	0.113	0.070	0.050	0.904	1.057
2	Mean	0.252	0.219	2.950	26.033	0.718	0.834
	Median	0.220	0.182	0.152	0.167	0.704	0.683
3	Mean	0.339	0.251	22.322	2.741	0.426	0.725
	Median	0.293	0.207	0.400	0.164	0.371	0.601
4	Mean	0.349	0.307	147.797	3.255	0.707	0.601
	Median	0.284	0.265	0.359	0.340	0.440	0.425
5	Mean	0.463	0.500	61.347	31557.498	1.445	1.836
	Median	0.364	0.402	0.658	0.350	0.239	0.482

where F is the Allocative Flexibility of category c , in region j and period t ; PPC is GDP per capita, HH is the region's Herfindahl–Hirschman index, and N is a measure of region size, i.e., the number of people composing its labor market. Regression results are located in [Table 22](#) and [Table 23](#), in [Chapter B](#).

The positive and significant coefficient for region size (N) indicates that larger regions, with greater economic diversity, have a greater capacity to absorb economic shocks, reflected in lower turbulence observed in these areas. Economic diversity associated with larger regions can indeed contribute to turbulence reduction due to greater capacity to absorb and mitigate negative impacts. However, a larger region may also increase mismatch, possibly due to greater variability between sectors.

The Hirschman Index (HH), measuring sectoral economic concentration, has a significant negative impact on flexibility. Regions with high sectoral concentration tend to be more sensitive to shocks in idiosyncratic sectors, leading to greater turbulence. High concentration can also increase mismatch, especially if the local economy is dominated by few sectors, hindering the reallocation of workers between distinct sectors. Therefore, dependence on a limited number of sectors can also restrict flexibility, making the region more vulnerable to changes or crises in specific sectors.

The positive and significant coefficient for GDP per capita (PPC) indicates that regions with higher per capita product levels are more flexible. They generally possess more robust and developed economies, providing more resources to deal with economic shocks and mitigate turbulence. Furthermore, higher GDP per capita is related to mismatch reduction, reflecting the presence of a greater diversity of high value-added sectors, offering more employment opportunities and adjusting the local economy to different economic conditions. Economic flexibility is also higher in regions with high GDP per capita, reflecting better market conditions and infrastructure facilitating adap-

tation to economic changes.

It is extremely important to observe the loss of statistical power of coefficients as portability groups become less portable.

It is also possible to identify some patterns related to important concepts in urban and regional economic studies.

Urbanization and localization economies are fundamental concepts in the study of urban and regional economics. Urbanization economies refer to benefits firms and individuals obtain by locating in large urban centers. These benefits include greater infrastructure availability, access to a diverse labor force, proximity to consumer markets and suppliers, and the presence of specialized services. Urbanization economies arise due to productivity increases resulting from the concentration of economic activities in a restricted geographical space.

On the other hand, localization economies are economic benefits resulting from the proximity of firms operating in the same sector or complementary sectors. [Marshall \(1890\)](#) was one of the first to identify these economies, highlighting three main factors: the creation of a specialized labor market, the availability of specialized suppliers, and knowledge diffusion among firms. These economies are particularly evident in industrial clusters, where geographical proximity facilitates cooperation and innovation.

[Ellison, Glaeser e Kerr \(2010\)](#) argues that both urbanization and localization economies play crucial roles in urban economic development, promoting innovation and competitiveness among firms. The interaction between these two forms of external economies can thus boost a region's economic growth, making it more attractive for new investments.

In summary, while urbanization economies are widely associated with scale and scope benefits provided by urban concentration, localization economies are more specific and related to firm proximity within similar or complementary sectors. Both are essential for understanding the dynamics of urban and regional economic development.

We identified a relationship between urbanization and localization economies. In general, occupations with high portability indices tend to take advantage of large city externalities (where diverse commerce and services prevail). Conversely, occupations with low portability indices benefit more from localization economies.

2.4 Conclusion

This study presented a comprehensive analysis of allocative flexibility in the Brazilian labor market, focusing on different regions of the country during periods of domestic crisis (2010–2016), economic recovery, and the COVID-19 pandemic (2016–2021). The investigation centered on occupational portability and the capacity of regions to adjust their labor markets in response to economic shocks.

The research revealed that poorer regions exhibit greater allocative flexibility, attributed to lower firm density in specific sectors, lower occupational complexity, and a greater presence of portable occupations. These factors allow workers to be reallocated more easily between different sectors, minimizing the negative impacts of economic shocks.

Crisis and pandemic periods were analyzed separately for a better understanding of regional dynamics. During the 2015–2016 economic crisis, shock sensitivity was more pronounced, especially in poorer regions, which demonstrated superior adjustment capacity. However, the COVID-19 pandemic presented distinct challenges, with high mismatch rates, particularly in regions with lower occupational portability. This suggests that while poorer regions may be more flexible during general economic crises, the specific nature of shocks and occupational complexity significantly affect this regional adaptive capacity.

The graphs and maps presented illustrated the spatial variation of allocative flexibility and occupational portability in Brazil. Regions with greater economic diversity and a higher number of portable occupations demonstrated greater resilience to shocks. In contrast, regions with high sectoral specialization faced greater challenges in worker reallocation, resulting in higher wage losses and prolonged unemployment periods.

This study contributes significantly to understanding allocative flexibility and occupational portability in Brazil. The results underscore the importance of regional economic diversity and adaptation to labor market changes. Policies promoting occupational portability and economic diversification can increase the resilience of Brazilian regions to economic shocks. The labor market's adjustment capacity is crucial for mitigating negative impacts of future crises and promoting sustainable and inclusive economic growth.

3 Wage Flexibility

3.1 Introduction, Literature Review, and Theoretical Approach

How wages adjust to economic shocks is a central issue in the analysis of labor market functioning. The flexibility or rigidity of wages directly influences adjustment mechanisms in the face of demand shocks, affecting unemployment levels, productivity, sectoral reallocation, and the effectiveness of monetary policy. In more flexible markets, changes in the unemployment rate are expected to trigger wage adjustments that restore equilibrium. Conversely, in contexts of greater rigidity, shocks may translate into persistent unemployment or misalignment between labor supply and demand.

From a theoretical standpoint, the neoclassical tradition considers that real wages are determined by the intersection of labor supply and demand, adjusting automatically to clear the market. In this model, unemployment tends to be transitory and voluntary, reflecting workers' intertemporal preferences. However, historical episodes such as the Great Depression, the stagflation of the 1970s, and the 2008 financial crisis revealed that nominal and real wages do not respond automatically to changes in the economic environment, paving the way for models that incorporate imperfections, bargaining, and rigidity.

Bargaining models, such as those based on the Nash solution, suggest that the observed wage is the result of a negotiation process between workers and employers, where bargaining power depends on local labor market conditions. In a typical arrangement, the bargaining objective function is given by:

$$\max_w [U(w) - U(b)]^\theta [\Pi(w) - \Pi_0]^{1-\theta}$$

where $U(w)$ represents the utility of the employed worker, $U(b)$ the utility of the unemployed worker (reservation wage or benefit), $\Pi(w)$ the employer's profit with the worker, Π_0 the profit in the absence of the worker, and θ the worker's bargaining power, which is negatively related to the unemployment rate. Since $\theta = \theta(u)$ with $\frac{d\theta}{du} < 0$, the derivative of the wage with respect to unemployment is negative:

$$\frac{\partial w}{\partial u} < 0 \quad \Rightarrow \quad \frac{\partial \ln w}{\partial \ln u} < 0$$

This expression formalizes the idea that in regions with higher unemployment, workers possess lower bargaining power and, therefore, accept lower wages. This is the theoretical core of the wage curve, proposed by [Blanchflower e Oswald \(1994\)](#) as an empirical alternative to the Phillips curve. While the latter relates wage *growth* to the unemployment rate at an aggregate level, the wage curve establishes a negative relationship between the *level* of real wages and the level of regional unemployment:

$$\frac{\partial \ln w}{\partial \ln u} = \alpha < 0 \quad (3.1)$$

The parameter α is interpreted as the elasticity of the real wage with respect to unemployment. Empirically, an elasticity of $\alpha = -0.1$ means that a 10% increase in the unemployment rate leads, on average, to a 1% reduction in real wages. This relationship has been observed in various countries and institutional contexts, suggesting the existence of a structural mechanism behind this behavior.

The international literature accumulates robust evidence of the existence of the wage curve. [Blanchflower e Oswald \(1994\)](#) used data from 16 OECD countries and found negative and statistically significant elasticities in most cases. In general, estimated values range between -0.05 and -0.15 , with greater sensitivity among youth, less skilled workers, and more flexible sectors. Subsequent works, such as [Blanchflower e Oswald \(2005\)](#), confirmed the robustness of this relationship even in economies with strong union presence or rigid labor institutions.

In developing countries, the evidence is also consistent. Studies in Latin America, such as [Pratap e Quintin \(2006\)](#), and in South Africa, such as [Kingdon e Knight \(2006\)](#), indicate similar elasticities, although with variations stemming from informality, segmentation, and the absence of formal bargaining mechanisms.

In Brazil, the literature on wage flexibility is more limited but growing. [Barros e Mendonça \(1997\)](#) analyzed PNAD data between 1982 and 1995 and found that the wage-unemployment elasticity is negative and statistically significant, but with considerable regional heterogeneity. Meanwhile, [Silva, Monsueto e Porsse \(2015\)](#), using PNAD data, showed that wage sensitivity to unemployment is higher among youth and in the commerce and services sectors, while it is lower in industry and the public sector.

A complementary strand to the wage flexibility literature concerns occupational self-selection. [Roy \(1951\)](#) proposes that individuals choose their occupation based on potential wages in different alternatives. The selection is made according to:

$$w = \begin{cases} w_1, & \text{if } w_1 > w_2 \\ w_2, & \text{if } w_2 > w_1 \end{cases} \Rightarrow w = \max\{w_1, w_2\}$$

This choice maximizes the worker's expected return but introduces sample selection bias in the estimates of observed wages, since the sample of workers in each occupation reflects only those for whom that occupation is more advantageous. This bias can be corrected with the two-step procedure proposed by [Heckman \(1979\)](#).

Despite the model's elegance, the practical application of the Heckman correction here requires an exogenous instrumental variable for selection, which is not always available. Furthermore, in the case of RAIS, which covers the universe of active formal jobs, the wage censoring problem is mitigated, as only employed workers are analy-

zed. Even so, self-selection may be present in the choice between sectors or types of occupation, which should be considered a limitation of the empirical model.

This work contributes to this debate by estimating the wage curve for Brazil between 2014 and 2022, using RAIS data combined with state unemployment rates. An innovation of the study lies in the segmentation of workers according to occupational portability, distinguishing between occupations with high and low portability. The central hypothesis is that wage flexibility differs between these groups: more portable occupations would be less sensitive to unemployment, given the greater dynamism of adjustment via reallocation.

The motivation for segmenting the wage curve by occupational groups is related to the concept of occupational portability, that is, the capacity of an occupation to be performed in multiple productive sectors. This notion is directly connected to the potential for worker reallocation in the face of specific shocks, functioning as an alternative adjustment channel beyond wages.

As argued by Darougheh (2024) (and in Chapter ??), occupations with high sectoral portability tend to have greater intersectoral transition capacity, affording them greater fluidity in the labor market. In contrast, highly specialized or sectorally concentrated occupations face greater frictions in reallocation, which may induce wage rigidity or persistent unemployment. Thus, the degree of occupational portability can act as a moderator of the wage-unemployment elasticity.

Formally, the portability of an occupation o can be represented by the dispersion measure of the sectoral allocation of its employment ties. The relative share of occupation o in sector s is defined as:

$$p_{os} = \frac{E_{os}}{\sum_s E_{os}} \quad (3.2)$$

where E_{os} represents the number of employment ties of occupation o in sector s . Based on this distribution, the broadness (portability) index of the occupation is calculated as:

$$m_o = 1 - \sum_s p_{os}^2 \quad (3.3)$$

Theory suggests that occupations with high portability allow the worker to move more easily between sectors in response to demand variations or specific productivity shocks. Consequently, adjustment to the business cycle may occur more via reallocation than via wage variations. On the other hand, in occupations of low portability (for example, highly technical roles or those associated with a single sector), wages tend to react more directly to changes in the unemployment level, since reallocation is limited.

This argument implies that the wage-unemployment elasticity, denoted by α , may differ between the two groups. Low portability occupations would tend to present a more elastic relationship:

$$|\alpha^{\text{low portability}}| > |\alpha^{\text{high portability}}| \quad (3.4)$$

If this differential is empirically confirmed, it will contribute to the literature on allocative and wage flexibility, highlighting that the intersectoral mobility of occupations influences the degree of wage rigidity or sensitivity to labor market conditions.

Thus, by estimating the wage curve separately for the two groups (workers in portable and non-portable occupations), the aim is not only to measure the general degree of flexibility of the Brazilian labor market but also to understand how this flexibility is conditioned by structural occupational attributes.

The next section describes the econometric methodology adopted, as well as the variables used, the controls applied, and the estimation strategy with panel data and fixed effects.

3.2 Methodology

The empirical strategy of this work seeks to estimate the elasticity of real wages with respect to the unemployment rate, in line with the classic wage curve literature ((BLANCHFLOWER; OSWALD, 1994);(BARROS; MENDONÇA, 1997)). The central objective is to analyze whether this elasticity varies between workers in occupations with different degrees of intersectoral portability, starting from the theoretical hypothesis that wage flexibility tends to be greater in markets where allocative flexibility is lower.

The basic econometric specification adopted is as follows:

$$\ln w_{ist} = \alpha \ln u_{st} + X'_{ist}\beta + \gamma_s + \delta_t + \varepsilon_{ist}, \quad (3.5)$$

where $\ln w_{ist}$ is the logarithm of the monthly real wage¹ of individual i , resident in state s and year t ; $\ln u_{st}$ is the logarithm of the state unemployment rate; X_{ist} represents a vector of observable individual characteristics (age, age squared, sex, race/color, among others); γ_s and δ_t are state and year fixed effects, respectively; and ε_{ist} is the error term.

The coefficient of interest is α , which captures the wage elasticity with respect to unemployment. Under wage bargaining theory, it is expected that $\alpha < 0$, indicating that real wages tend to fall when unemployment rises. This sensitivity, however, may vary between occupational portability groups, defined based on the occupational dispersion index m_o . Occupations with an index greater than 0.91 are considered portable.

Regarding the possibility of sample selection bias, the literature frequently recommends applying the Heckman correction (HECKMAN, 1979) to deal with truncated or censored samples, which could cause bias in estimation if only observed wages are used. Formally, the selection model consists of two stages:

$$\text{Selection equation: } D_i^* = Z_i'\pi + v_i, \quad D_i = 1 \text{ if } D_i^* > 0, \text{ otherwise } 0, \quad (3.6)$$

$$\text{Structural equation: } y_i = X_i'\beta + u_i, \quad (3.7)$$

where D_i indicates the observation of wage y_i and Z_i contains variables that influence selection. The Heckman correction seeks to estimate the conditional expectation of the error term given selection:

$$\mathbb{E}[u_i|D_i = 1] = \rho\sigma_u\lambda(Z_i'\pi), \quad (3.8)$$

where $\lambda(\cdot)$ is the Inverse Mills Ratio, and ρ represents the correlation between u_i and v_i .

However, in the present study, the Heckman correction was not implemented for two main reasons. First, as previously mentioned, the database used (RAIS) contains

¹ The data were deflated using the Broad Consumer Price Index (IPCA) of the period.

only active formal ties, meaning there is no direct censoring or absence of observable wages for employed individuals, eliminating the typical truncated sampling problem. Second, despite possible self-selection in occupational choice, which could cause bias if correlated with the wage regression error, there are no strong and plausible instrumental variables for the occupational decision that allow for an adequate identification of the two-stage model. Thus, applying the Heckman correction would be vulnerable to a lack of identification and low robustness of results.

Finally, the expected results should reflect the theoretical hypothesis: low portability occupations present a higher wage elasticity in magnitude (larger $|\alpha|$), due to workers' lower reallocation capacity in response to adverse shocks in the labor market. Conversely, high portability occupations present lower elasticities, as the adjustment to shocks occurs partly via allocative flexibility, attenuating the need for wage adjustments. This differentiation can be formalized as:

$$\alpha = \begin{cases} \alpha_L, & \text{if } m_o \leq 0.91 \quad (\text{low portability}), \\ \alpha_H, & \text{if } m_o > 0.91 \quad (\text{high portability}), \end{cases} \quad \text{with } |\alpha_L| > |\alpha_H|. \quad (3.9)$$

3.3 Data and Results

3.3.1 Descriptive Statistics

Data related to workers are from RAIS, and data related to unemployment by state are from the Continuous PNAD. Table 10 presents descriptive statistics of formal workers between 2014 and 2022. A slight increase in the average age of workers over the period is observed, rising from 34.8 years in 2014 to 36.5 years in 2022. The proportion of women in formal employment remained relatively stable, with a slight rise from 39.5% to 41.0%.

Regarding real wages, there was growth in both the mean and median over the analyzed period. The median wage practically doubled, going from R\$ 794.78 in 2014 to R\$ 2,000.00 in 2022, reflecting the cumulative effects of inflation, wage adjustments, and post-crisis recomposition. The disparity between mean and median, as well as the high standard deviation (notably in 2022), indicate significant heterogeneity in the wage distribution.

Tabela 10 – Descriptive statistics by year

Year	n (mi)	Mean Age	Med. Age	Mean W	Med. W	SD	% women
2014	38.52	34.8	33	1,232.04	794.78	1,650.72	39.5
2015	37.10	35.3	34	1,477.70	957.43	1,982.17	39.9
2016	35.46	35.6	34	1,685.93	1,106.33	2,209.11	40.3
2017	35.33	35.8	34	1,827.84	1,207.99	2,364.04	40.3
2018	35.84	36.0	35	1,938.74	1,286.23	2,500.02	40.3
2019	35.20	36.1	35	2,092.52	1,392.40	2,699.88	40.5
2020	34.87	36.3	35	2,172.74	1,436.52	2,913.23	40.0
2021	37.38	36.3	35	2,562.81	1,696.81	3,398.22	40.6
2022	39.51	36.5	35	3,664.75	2,000.00	13,655.86	41.0

3.3.2 Regression Results

Below are the results of the estimation of equation (3.5), separately for the high and low occupational portability groups. The models include state and year fixed effects, and standard errors are robust and clustered by federation unit.

The estimated coefficients for the logarithm of the unemployment rate ($\ln u_{st}$) are negative and statistically significant in both groups, empirically confirming the presence of a wage curve: increases in unemployment are associated with wage reductions.

Notably, the effect is more pronounced among workers with low occupational portability. This result corroborates the work's hypothesis. That is, when allocative flexibility is limited, wages adjust more intensely to adverse shocks in the labor market, such as increases in unemployment.

Dependent variable: $\ln_wage$		
Model:	(1) High Portability	(2) Low Portability
$\ln_unemployment$	-0.0544*** (0.0127)	-0.0603*** (0.0201)
Observations	35,940,718	292,905,906
R^2	0.3971	0.2916

Standard errors clustered by state in parentheses.

*** $p < 0.01$

3.4 Conclusion

This small and simple study sought to investigate wage flexibility in the Brazilian formal labor market between 2014 and 2022, in light of the wage curve literature and bargaining theory. Starting from the hypothesis that wages react negatively to variations in the unemployment rate, the wage-unemployment elasticity was estimated based on RAIS microdata and state unemployment rates, distinguishing between occupations with high and low intersectoral portability.

The empirical results confirm the existence of a negative relationship between real wages and unemployment, in line with international and national literature. More importantly, it was verified that this elasticity is heterogeneous among occupational groups: workers in low portability occupations present greater wage sensitivity to the economic cycle, while those in more portable occupations tend to have more stable wages in the face of unemployment fluctuations.

This evidence supports the hypothesis that allocative flexibility acts as a moderator of wage flexibility. In portable occupations, adjustment to economic shocks occurs more intensely via reallocation between sectors, reducing the need for wage adjustments. In contrast, in occupations that are more restricted in sectoral terms, the wage becomes the main channel of adaptation, resulting in a steeper wage curve.

These findings have relevant implications for public policy formulation. First, they reinforce the importance of professional qualification and retraining policies that expand the portability of occupations, especially in contexts of technological transformations and sectoral shocks. Second, they suggest that the redistributive and welfare effects of economic fluctuations are not uniform, being more severe for workers in low-mobility occupations. Finally, they indicate that the effectiveness of monetary policy and unemployment dynamics depends, in part, on the occupational structure of the economy.

Limitations include the absence of formal correction mechanisms for possible occupational self-selection, as well as the impossibility of incorporating informal ties, which restricts the scope of results to the formal segment of the labor market. Future research could explore these dimensions, as well as analyze interactions between wage flexibility, informality, and employment protection policies.

In summary, this work contributes to the literature by showing that the elasticity of wages to unemployment is not only a macroeconomic or institutional characteristic but also depends on structural attributes of occupations. By combining the concepts of wage bargaining and occupational portability, it offers a new lens for understanding the adjustment mechanisms of the Brazilian labor market.

Referências

- ABRAHAM, K. G.; KATZ, L. F. Cyclical unemployment: sectoral shifts or aggregate disturbances? *Journal of political Economy*, The University of Chicago Press, v. 94, n. 3, Part 1, p. 507–522, 1986.
- ASSUNÇÃO, L. M. C. d. M.; RIBEIRO, L. C. d. S.; MOURA, F. R. d. Efeitos do isolamento social sobre o setor de serviços do brasil: uma análise a partir do modelo varx. *Nova Economia*, SciELO Brasil, v. 34, p. e8339, 2024.
- AUTOR, D. H. The “task approach” to labor markets: an overview. *Journal for Labour Market Research*, SpringerOpen, v. 46, n. 3, p. 185–199, 2013.
- AUTOR, D. H.; LEVY, F.; MURNANE, R. J. The skill content of recent technological change: An empirical exploration. *The Quarterly journal of economics*, MIT Press, v. 118, n. 4, p. 1279–1333, 2003.
- BARROS, R. P. d.; MENDONÇA, R. S. P. d. Flexibilidade do mercado de trabalho brasileiro: uma avaliação empírica. Instituto de Pesquisa Econômica Aplicada (Ipea), 1997.
- BARROS, R. P. de et al. Uma avaliação empírica do grau de flexibilidade alocativa do mercado de trabalho brasileiro. *Planejamento e Políticas Públicas*, n. 15, 1997.
- BLANCHFLOWER, D. G.; OSWALD, A. J. *The wage curve*. [S.l.]: MIT press, 1994.
- BLANCHFLOWER, D. G.; OSWALD, A. J. *The wage curve reloaded*. [S.l.]: National Bureau of Economic Research Cambridge, Mass., USA, 2005.
- CASTRO, G. H. L. d. Habilidades dos trabalhadores na indústria brasileira. 2023.
- DAROUGHIEH, S. Occupation-industry mismatch in the cross section and the aggregate. *Journal of Political Economy Macroeconomics*, The University of Chicago Press Chicago, IL, v. 2, n. 3, p. 375–408, 2024.
- DEMING, D. J. The growing importance of social skills in the labor market. *The quarterly journal of economics*, Oxford University Press, v. 132, n. 4, p. 1593–1640, 2017.
- ELLISON, G.; GLAESER, E. L.; KERR, W. R. What causes industry agglomeration? evidence from coagglomeration patterns. *American Economic Review*, American Economic Association, v. 100, n. 3, p. 1195–1213, 2010.
- FAGGIO, G.; SALVANES, K. G.; REENEN, J. V. The evolution of inequality in productivity and wages: panel data evidence. *Industrial and Corporate Change*, Oxford University Press, v. 19, n. 6, p. 1919–1951, 2010.
- FARINHA, T. et al. What drives the geography of jobs in the us? unpacking relatedness. *Industry and Innovation*, Taylor & Francis, v. 26, n. 9, p. 988–1022, 2019.
- FERREIRA, B. et al. Ocupações agrícolas e não agrícolas: trajetória e rendimentos no meio rural brasileiro. 2006.

FRENKEN, K.; OORT, F. V.; VERBURG, T. Related variety, unrelated variety and regional economic growth. *Regional studies*, Taylor & Francis, v. 41, n. 5, p. 685–697, 2007.

GATHMANN, C.; SCHÖNBERG, U. How general is human capital? a task-based approach. *Journal of Labor Economics*, The University of Chicago Press, v. 28, n. 1, p. 1–49, 2010.

GIOVANINI, A.; PEREIRA, W. M.; ALMEIDA, H. J. F. Diversidade produtiva e crescimento econômico: algumas evidências para os municípios brasileiros. *Nova Economia*, SciELO Brasil, v. 32, n. 03, p. 687–717, 2022.

HECKMAN, J. J. Sample selection bias as a specification error. *Econometrica: Journal of the econometric society*, JSTOR, p. 153–161, 1979.

JACINTO, P. d. A.; RIBEIRO, E. P. Crescimento da produtividade no setor de serviços e da indústria no brasil: dinâmica e heterogeneidade. *Economia Aplicada*, SciELO Brasil, v. 19, n. 3, p. 401–427, 2015.

JACQUEMIN, A. P.; BERRY, C. H. Entropy measure of diversification and corporate growth. *The journal of industrial economics*, JSTOR, v. 27, n. 4, p. 359–369, 1979.

JARA-FIGUEROA, C. et al. The role of industry-specific, occupation-specific, and location-specific knowledge in the growth and survival of new firms. *Proceedings of the National Academy of Sciences*, National Academy of Sciences, v. 115, n. 50, p. 12646–12653, 2018.

JÚNIOR, E. C. d. N.; AZZONI, C. R.; CHAGAS, A. S. Habilidades e prêmio salarial urbano. Instituto de Pesquisa Econômica Aplicada (Ipea), 2021.

JÚNIOR, E. C. d. N. N. Habilidades, mudanças de firma e prêmio salarial urbano. 2018.

KINGDON, G. G.; KNIGHT, J. How flexible are wages in response to local unemployment in south africa? *ILR Review*, SAGE Publications Sage CA: Los Angeles, CA, v. 59, n. 3, p. 471–495, 2006.

LAYARD, P. R. G. et al. *Unemployment: macroeconomic performance and the labour market*. [S.l.]: Oxford University Press, USA, 2005.

LILIEN, D. M. Sectoral shifts and cyclical unemployment. *Journal of political economy*, The University of Chicago Press, v. 90, n. 4, p. 777–793, 1982.

LONGHI, S.; TAYLOR, M. *Employed and unemployed job seekers: Are they substitutes?* [S.l.], 2011.

LONGHI, S.; TAYLOR, M. *Explaining differences in job search outcomes between employed and unemployed job seekers*. [S.l.], 2011.

LONGHI, S.; TAYLOR, M. Occupational change and mobility among employed and unemployed job seekers. *Scottish Journal of Political Economy*, Wiley Online Library, v. 60, n. 1, p. 71–100, 2013.

MACEDO, F. C. d.; PORTO, L. R. Aspectos da evolução regional do mercado de trabalho formal no brasil (2002-2018). Instituto de Pesquisa Econômica Aplicada (Ipea), 2023.

MACIENTE, A. N. *The determinants of agglomeration in Brazil: input-output, labor and knowledge externalities*. [S.l.]: University of Illinois at Urbana-Champaign, 2013.

MACIENTE, A. N.; RAUEN, C. V.; KUBOTA, L. C. Tecnologias digitais, habilidades ocupacionais e emprego formal no brasil entre 2003 e 2017. Instituto de Pesquisa Econômica Aplicada (Ipea), 2019.

MARIONI, L. d. S. Overeducation in the labour market: evidence from brazil. *Education Economics*, Taylor & Francis, v. 29, n. 1, p. 53–72, 2021.

MARSHALL, A. 1920. principles of economics. *London: Mac-Millan*, p. 1–627, 1890.

MORCEIRO, P. C. A indústria brasileira no limiar do século xxi: uma análise da sua evolução estrutural, comercial e tecnológica. *Universidade de São Paulo*, 2018.

NEFFKE, F. et al. The dynamics of agglomeration externalities along the life cycle of industries. *Regional studies*, Taylor & Francis, v. 45, n. 1, p. 49–65, 2011.

OREIRO, J. L.; FEIJÓ, C. A. Desindustrialização: conceituação, causas, efeitos e o caso brasileiro. *Brazilian Journal of Political Economy*, SciELO Brasil, v. 30, p. 219–232, 2010.

PISSARIDES, C. A. *Equilibrium unemployment theory*. [S.l.]: MIT press, 2000.

PRATAP, S.; QUINTIN, E. Are labor markets segmented in developing countries? a semiparametric approach. *European Economic Review*, Elsevier, v. 50, n. 7, p. 1817–1841, 2006.

RODRIGUES-SILVEIRA, R. *Structural changes in Brazilian employment (2002-2021)*. [S.l.], 2023.

ROY, A. D. Some thoughts on the distribution of earnings. *Oxford economic papers*, Oxford University Press, v. 3, n. 2, p. 135–146, 1951.

SABOIA, J. Descentralização industrial no brasil na década de noventa: um processo dinâmico e diferenciado regionalmente. *Nova Economia*, Departamento de Ciências Econômicas, v. 11, n. 2, p. 85–122, 2001.

SABOIA, J. A continuidade do processo de desconcentração regional da indústria brasileira nos anos 2000. *Nova economia*, SciELO Brasil, v. 23, p. 219–278, 2013.

SABOIA, J. L. M. Desconcentração industrial no brasil nos anos 90: um enfoque regional. *Texto para discussão*, Universidade Federal do Rio de Janeiro, 2000.

SANTOS, M. F. et al. Mercado de trabalho em crise: uma análise da resiliência do emprego formal do estado de minas gerais. Universidade Federal de Minas Gerais, 2023.

SHANNON, C. E. A mathematical theory of communication. *The Bell system technical journal*, Nokia Bell Labs, v. 27, n. 3, p. 379–423, 1948.

SHAPIRO, C.; STIGLITZ, J. E. Equilibrium unemployment as a worker discipline device. *The American economic review*, JSTOR, v. 74, n. 3, p. 433–444, 1984.

SILVA, A. S. P. d.; MONSUETO, S. E.; PORSSE, A. A. Flexibilidade do mercado de trabalho: uma análise comparativa entre segmentos socioeconômicos no brasil (2002-2009). Instituto de Pesquisa Econômica Aplicada (Ipea), 2015.

SILVA, F. A. G. d.; MONTE, P. A. d. Incompatibilidade escolaridade-ocupação no brasil: incidência de sobre-educação de mão de obra ou escassez de empregos qualificados? *Revista de Economia Contemporânea*, SciELO Brasil, v. 28, p. e242806, 2024.

STADUTO, J. A. R.; SCHIO, T. A. Flexibilidade alocativa do mercado de trabalho brasileiro: uma avaliação empírica dos estados e das regiões metropolitanas e não-metropolitanas. *Revista Brasileira de Estudos Regionais e Urbanos*, v. 4, n. 2, p. 51-66, 2010.

Apêndices

APÊNDICE A – Figures

Figura 15 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region); Total

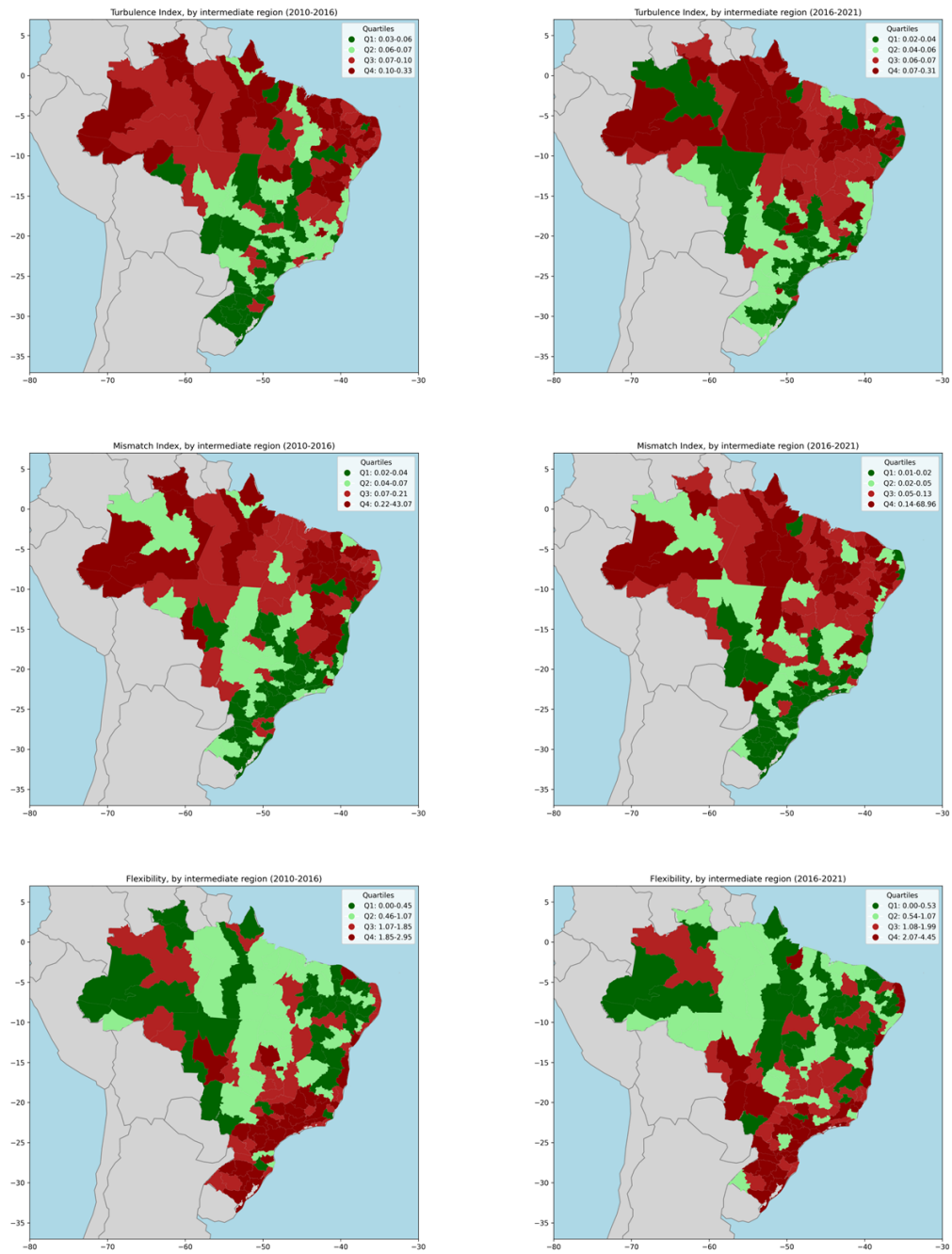


Figura 16 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region);
Category 1

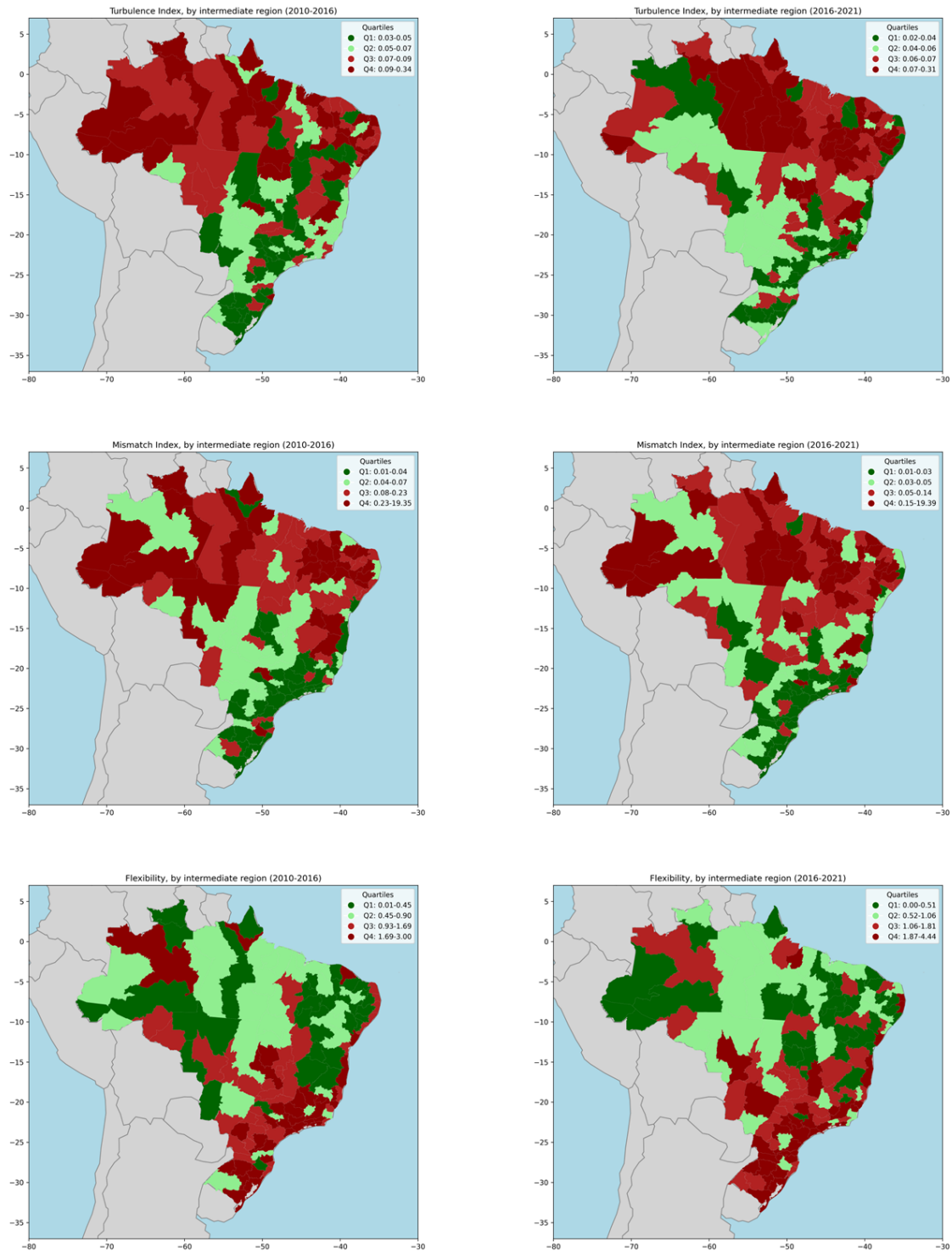


Figura 17 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region);
Category 2

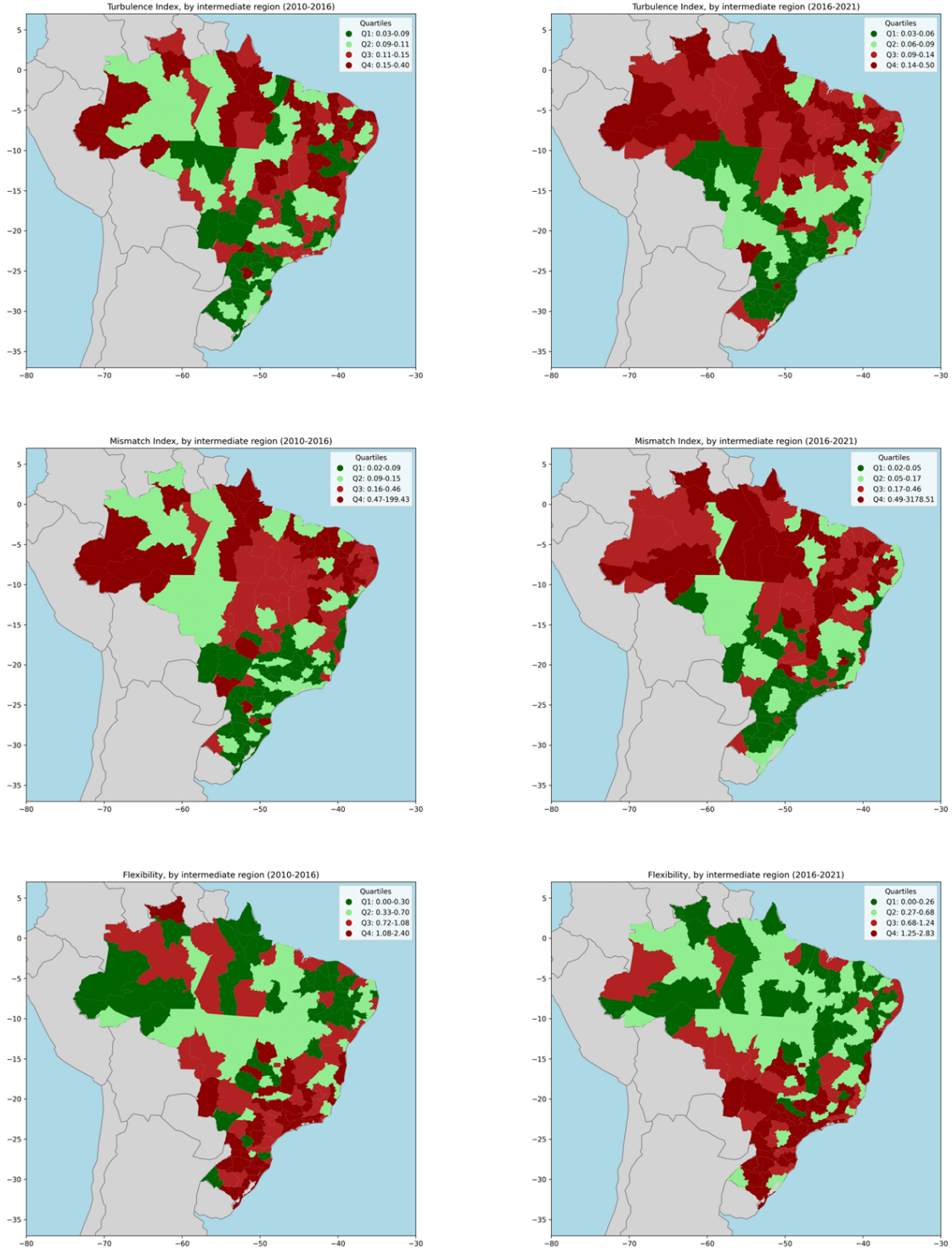


Figura 18 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region);
Category 3

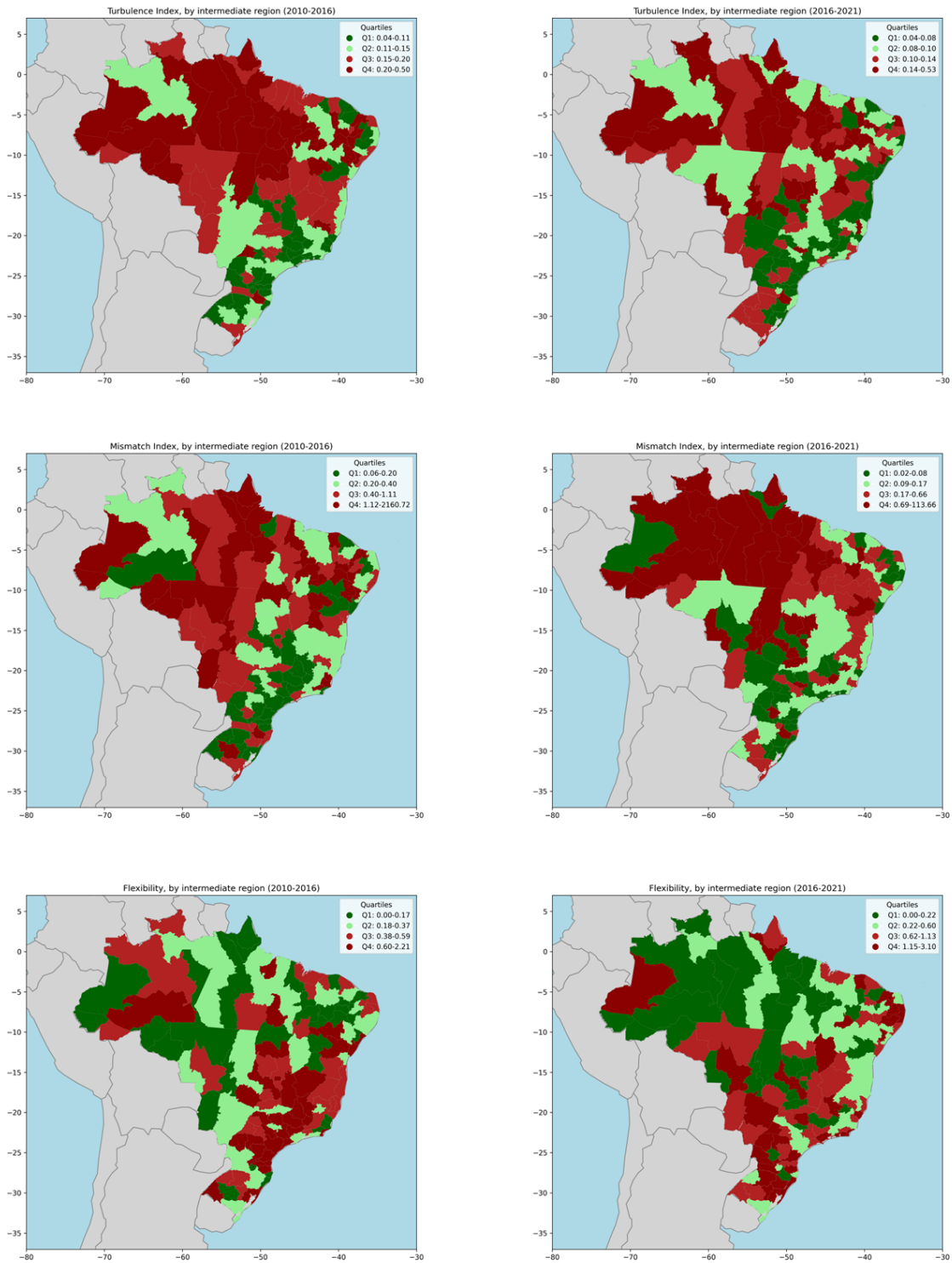


Figura 19 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region);
Category 4

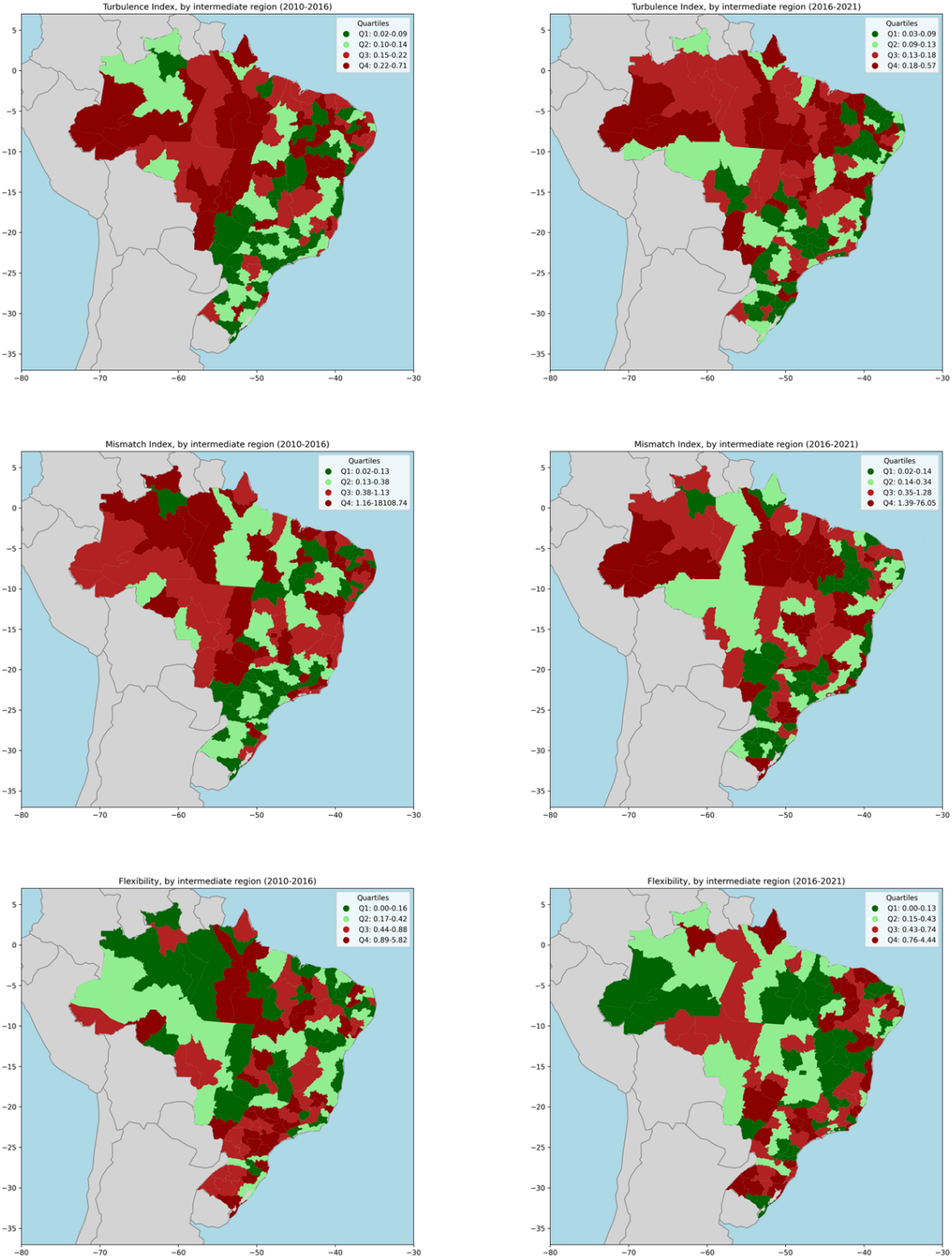
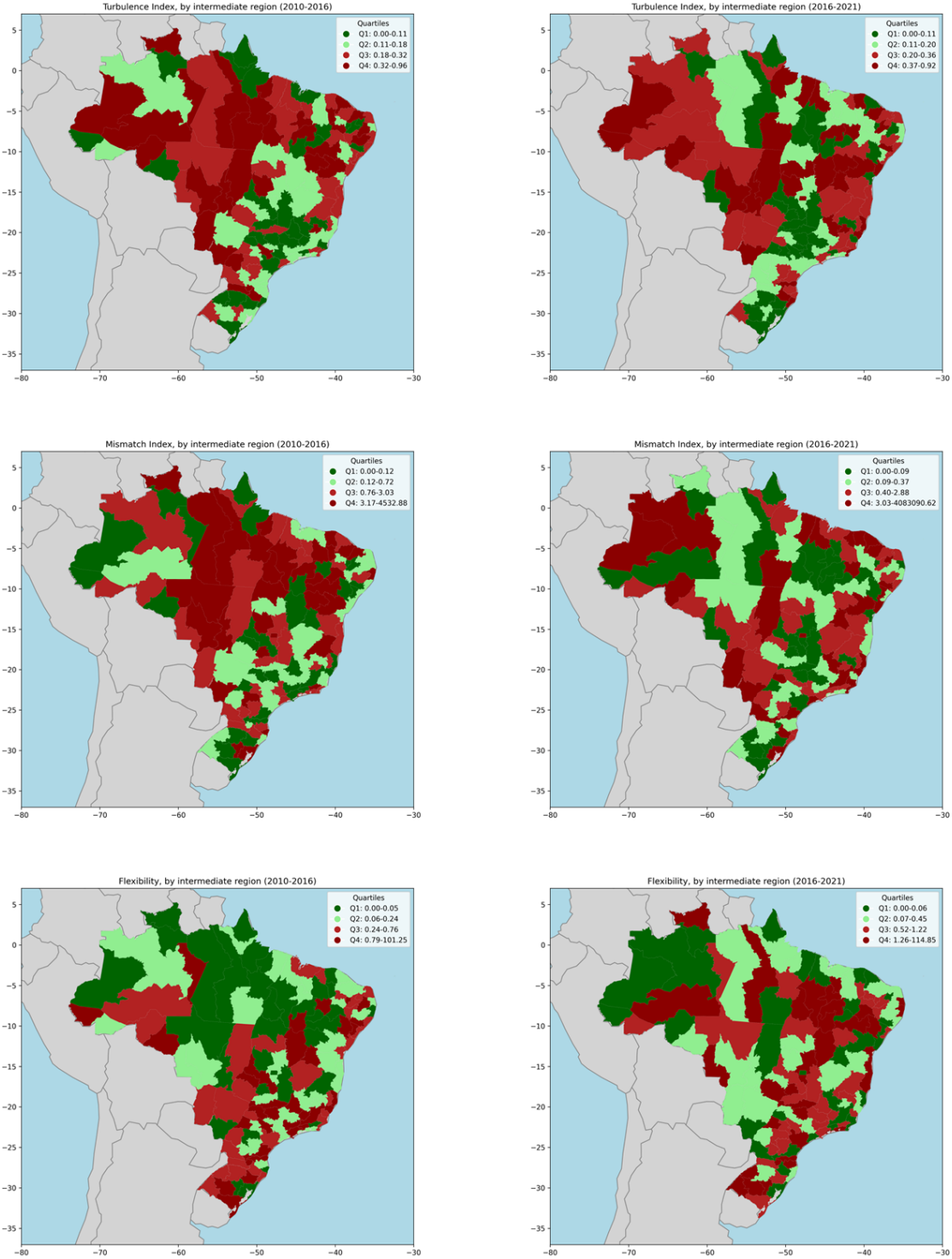


Figura 20 – Turbulence, Mismatch, and Allocative Flexibility (by intermediate region);
Category 5



APÊNDICE B – Tables

Tabela 11 – Percentage distribution of the employed labor force by occupational group (2010–2023).

Year	0	1	2	3	4	5	6	7	8	9
2010	0.0%	3.3%	6.4%	8.2%	18.9%	25.3%	4.4%	26.0%	4.4%	2.9%
2011	0.0%	3.4%	6.7%	8.4%	19.0%	25.3%	4.3%	25.7%	4.3%	2.9%
2012	0.0%	3.5%	6.9%	8.5%	19.2%	25.5%	4.0%	25.4%	4.1%	2.9%
2013	0.0%	3.7%	7.0%	8.6%	19.3%	25.6%	3.9%	24.9%	4.2%	2.8%
2014	0.0%	3.8%	7.1%	8.8%	19.5%	26.1%	3.7%	24.0%	4.1%	2.8%
2015	0.0%	3.9%	7.4%	8.9%	19.7%	26.8%	3.8%	22.5%	4.1%	2.8%
2016	0.0%	4.0%	7.6%	9.1%	19.7%	27.3%	3.9%	21.4%	4.1%	2.8%
2017	0.0%	4.0%	7.8%	9.2%	19.7%	27.5%	4.0%	21.0%	4.2%	2.7%
2018	0.0%	4.0%	7.9%	9.3%	19.7%	27.7%	3.9%	20.7%	4.0%	2.7%
2019	0.0%	4.1%	8.0%	9.3%	19.6%	27.8%	3.9%	20.6%	4.0%	2.7%
2020	0.0%	4.1%	8.2%	9.5%	19.5%	26.8%	3.8%	21.3%	4.2%	2.7%
2021	0.0%	4.2%	8.3%	9.6%	19.4%	26.5%	3.8%	21.3%	4.0%	2.7%
2022	0.0%	4.3%	8.4%	9.5%	18.9%	26.9%	3.9%	21.2%	3.9%	3.0%
2023	0.0%	4.4%	8.4%	9.6%	18.9%	27.1%	3.8%	21.2%	3.9%	2.7%

Source: author's elaboration with RAIS-MTE data. *Note:* 0 — Members of the Armed Forces, Police, and Military Firefighters; 1 — Senior Members of Public Power, Leaders of Organizations of Public Interest and Companies, Managers; 2 — Professionals of Sciences and Arts; 3 — Mid-Level Technicians; 4 — Administrative Services Workers; 5 — Services Workers, Shop and Market Sales Workers; 6 — Agricultural, Forestry, and Fishery Workers; 7 — Workers in the Production of Industrial Goods and Services; 8 — Workers in the Production of Industrial Goods and Services; 9 — Repair and Maintenance Services Workers.

Tabela 12 – Occupations comprising the most numerous 1% of total occupations classified in the CBO, by formal private ties (2010–2023)

Occupation	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Retail salesperson	2052934	2138841	2211663	2257298	2256483	2123464	2005654	1940071	1899661	1874001	1751380	1815739	1933912	1945554
Office clerk	1674145	1737786	1801191	1816145	1814394	1718495	1628384	1578962	1547381	1525130	1435800	1489607	1566229	1578366
Cleaner	765792	832616	957529	1043066	1128731	1187095	1216922	1279466	1384614	1417792	1420371	1514517	1712894	1798004
Administrative assistant	981313	1067667	1115424	1181831	1224023	1201102	1179506	1171784	1200843	1228346	1219433	1329843	1460803	1550046
Production line feeder	870424	907306	939985	967252	947991	880460	857138	908823	956159	980315	1095889	1153948	1230147	1267722
Truck driver	737286	799962	849117	897712	919916	893742	848395	817522	823034	837098	880191	943380	1055085	1118979
Cashier	649676	700359	751637	800135	835494	851052	822657	845188	866423	869104	867920	904275	963371	985003
Construction laborer	896163	932280	923209	899442	821609	668966	520955	473227	464432	468169	497480	544188	636925	664437
Security guard	528877	568277	624283	633262	644362	617480	575275	534214	550468	543743	539278	542248	576657	595567
Building doorman	440116	471315	493622	512620	536380	551274	548274	552776	563151	558153	563618	564917	606856	623843
Receptionist, general	434935	469177	495438	512442	522907	514202	497781	500905	509866	513743	485887	507818	557322	577570
Stock clerk	358583	377670	405819	426500	445328	444709	470074	477082	504323	529282	549419	567845	584724	598877
Cook, general	456847	485192	496653	509271	520234	496557	465703	450799	433737	424889	355283	384171	456629	479758
Nursing technician	222778	257716	300126	329926	369442	384955	396873	413551	436516	459540	518552	557835	576638	586415
Storekeeper	356533	391233	413427	436565	449245	430786	421138	404289	391171	377410	375783	392676	405848	408615
Mason	415528	454404	468984	469436	452731	385911	317302	282136	273594	268846	270567	295984	356642	381391
Agricultural worker, general	370982	370995	356764	355620	354497	354631	352963	353636	346422	336905	324589	331242	379043	378250
Snack bar attendant	316865	339915	352064	379640	389398	375206	366644	370652	367466	372889	308267	324915	385559	406753
Cleaning service worker	462258	453764	426285	416068	405886	379767	343724	313364	286824	271007	258094	261062	290058	300019
Administrative supervisor	258881	282437	298388	325944	317775	319130	312555	305613	308765	308449	309808	321118	345526	366618
Driver's helper	262268	278274	285513	289895	299054	279149	265597	265964	272751	277872	280065	287241	306596	315640
Administrative manager	250298	277226	297730	315776	325764	325773	318352	312615	307572	299661	294296	310780	336730	344153
Gas station attendant	-	223725	234596	244624	254874	261239	256425	258385	258642	259479	252801	266531	283354	298613
Shop and market attendant	-	-	-	-	-	-	-	252982	341893	392522	413050	489071	603155	674730
Food service assistant	-	-	-	-	-	237229	265014	296725	348887	370029	328738	368603	448859	499614
Telemarketing operator	-	-	-	-	-	299080	290151	291709	292590	291771	316114	345689	327782	308759
Urban bus driver	224429	235407	244956	247378	255558	254831	245152	-	-	-	215399	-	-	-
Watchman	245263	253550	240650	239914	233918	-	-	-	-	-	-	-	-	-
Sugarcane worker	260239	252723	-	-	-	-	-	-	-	-	-	-	-	-
Administrator	-	-	-	-	-	-	-	-	-	-	-	222175	250625	261954
Building caretaker	-	-	-	234787	236990	-	-	-	-	-	-	-	-	-
Total	14493413	15559817	15985053	16742549	16962984	16436285	15788608	15652440	15937185	16056145	16128072	17037418	18637969	19315250

Note: Missing values are indicated by dashes (-) and correspond to occupations not classified in the respective year.

Source: author's elaboration with RAIS-MTE data

Tabela 13 – Total occupations by selected sector (2010–2023)

Sector	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Retail trade	5997484	6308979	6589552	6811844	6965954	6849069	6650428	6619667	6596201	6549775	6411750	6702336	7057157	7119625
Land transport	1470130	1575610	1636717	1715560	1770547	1733529	1651865	1628226	1648411	1660140	1619349	1670587	1820502	1913750
Food product manuf.	1399365	1451614	1438865	1504364	1531271	1529077	1478725	1526472	1522540	1577659	1634504	1670368	1805837	1867302
Wholesale trade, except motor vehicles	1378104	1452537	1527339	1582921	1639265	1616116	1597800	1606392	1622836	1667054	1681435	1797885	1952190	2058237
Human health activities	1322613	1436534	1537378	1633031	1726544	1769045	1793372	1823149	1890010	1951005	2053974	2215024	2365117	2466914
Agriculture, livestock, etc.	1295710	1360134	1343333	1357066	1355212	1376703	1354662	1379471	–	–	1338411	–	–	–
Building services	1265604	1381199	1409867	1472908	1537682	1542815	1537879	1525746	1591349	1541197	1531751	1552639	1698706	1714804
Education	1259043	1338424	1424154	1499949	1598521	1635998	1629715	1652851	1714079	1700595	1603858	1620150	1834128	1900356
Food services	1237129	1333529	1417369	1502347	1583938	1578274	1547192	1555411	1582846	1597581	–	1397197	1686594	1799854
Building construction	1189712	1302681	1335022	1347716	–	–	–	–	–	–	–	–	–	–
Office, admin support, other business services	–	–	–	–	–	–	–	–	–	1386296	1408738	1478645	1621911	1831955
													2030197	

Source: author's elaboration with RAIS-MTE data. *Note:* The table shows how many sectors concentrate half of the formal labor force in each year. Missing values correspond to sectors that are not present in the top 50% in the respective year.

Tabela 14 – Shannon Entropy by macro-region (2010–2023)

Year	North	Northeast	Southeast	South	Center-West
2010	3.439	3.480	3.658	3.592	3.377
2011	3.450	3.477	3.659	3.593	3.396
2012	3.429	3.478	3.651	3.598	3.411
2013	3.416	3.474	3.640	3.607	3.411
2014	3.404	3.461	3.626	3.581	3.410
2015	3.382	3.445	3.606	3.572	3.395
2016	3.367	3.434	3.589	3.561	3.387
2017	3.377	3.430	3.579	3.555	3.383
2018	3.389	3.440	3.580	3.562	3.380
2019	3.389	3.441	3.592	3.567	3.375
2020	3.399	3.448	3.601	3.572	3.377
2021	3.411	3.458	3.611	3.587	3.402
2022	3.425	3.479	3.623	3.600	3.419
2023	3.442	3.478	3.626	3.604	3.428

Source: author's elaboration.

Tabela 15 – Average portability by macro-region (2010–2023)

Year	North	Northeast	Southeast	South	Center-West
2010	0.6682	0.6509	0.6816	0.6631	0.6515
2011	0.6684	0.6529	0.6808	0.6664	0.6547
2012	0.6644	0.6515	0.6815	0.6698	0.6571
2013	0.6602	0.6492	0.6781	0.6645	0.6567
2014	0.6557	0.6447	0.6743	0.6639	0.6518
2015	0.6493	0.6351	0.6665	0.6600	0.6450
2016	0.6447	0.6293	0.6616	0.6573	0.6407
2017	0.6447	0.6280	0.6593	0.6546	0.6382
2018	0.6427	0.6254	0.6587	0.6552	0.6367
2019	0.6463	0.6277	0.6626	0.6579	0.6362
2020	0.6508	0.6323	0.6695	0.6660	0.6416
2021	0.6556	0.6359	0.6745	0.6713	0.6472
2022	0.6597	0.6416	0.6791	0.6755	0.6530
2023	0.6650	0.6473	0.6840	0.6794	0.6585

Source: author's elaboration.

Tabela 16 – Turbulence Index in Intermediate Regions by Macro-region (2010–2023)

Year	North	Northeast	Southeast	South	Center-West
2010	0.076	0.066	0.038	0.033	0.047
2011	0.091	0.065	0.033	0.029	0.047
2012	0.090	0.052	0.035	0.030	0.044
2013	0.075	0.055	0.033	0.029	0.044
2014	0.065	0.052	0.030	0.027	0.037
2015	0.064	0.057	0.035	0.027	0.036
2016	0.066	0.051	0.028	0.027	0.034
2017	0.067	0.043	0.028	0.026	0.031
2018	0.062	0.047	0.028	0.026	0.036
2019	0.062	0.055	0.033	0.028	0.039
2020	0.062	0.056	0.034	0.034	0.034
2021	0.056	0.052	0.029	0.026	0.033
2022	0.084	0.062	0.036	0.032	0.047
2023	0.054	0.041	0.026	0.025	0.033

Tabela 17 – Mismatch Index in Intermediate Regions by Macro-region (2010–2023)

Year	North	Northeast	Southeast	South	Center-West
2010	0.687	0.164	0.032	0.021	0.089
2011	0.553	0.160	0.053	0.015	0.257
2012	0.357	0.089	0.039	0.017	0.080
2013	0.255	0.913	0.023	0.165	0.048
2014	0.124	0.128	0.018	0.011	0.051
2015	0.054	0.092	0.019	0.031	0.029
2016	0.062	0.107	0.015	0.022	0.039
2017	0.085	0.153	0.020	0.016	0.021
2018	0.093	0.294	0.014	0.010	0.028
2019	0.075	0.222	0.028	0.014	0.033
2020	0.409	0.961	0.032	0.014	0.015
2021	0.124	0.066	0.016	0.010	0.030
2022	0.345	0.202	0.023	0.014	0.093
2023	0.080	0.043	0.011	0.013	0.017

Tabela 18 – Allocative Flexibility Index in Intermediate Regions by Macro-region (2010–2023)

Year	North	Northeast	Southeast	South	Center-West
2010	0.836	0.930	1.889	2.068	1.330
2011	0.743	0.894	2.062	2.474	1.143
2012	0.941	1.322	2.532	2.711	1.448
2013	1.238	1.186	2.536	2.513	1.508
2014	1.422	1.256	2.728	3.652	1.630
2015	1.696	1.454	3.183	3.416	2.608
2016	1.720	1.902	3.270	3.747	2.079
2017	1.241	1.520	2.861	3.545	1.820
2018	1.180	1.290	2.932	3.498	2.091
2019	1.344	1.335	2.349	2.788	1.808
2020	1.346	1.332	2.812	2.997	2.601
2021	1.210	1.442	2.938	3.416	1.868
2022	0.844	1.072	2.524	3.016	1.286
2023	1.295	1.789	3.492	3.446	2.803

Tabela 19 – Largest and smallest percentage growth in regional formal employment by sector (2010–2016)

Sector	Variation (%)
<i>Largest growth</i>	
Veterinary activities	112.78
IT services	35.41
Social assistance (w/o accom.)	44.97
Legal, accounting, and auditing	41.20
Human health	35.59
Real estate activities	39.62
Telecommunications	27.14
Water transport	27.60
Food services	25.06
Maintenance and repair of machinery	20.83
<i>Largest declines</i>	
Sewerage and related	-48.28
Gambling and betting	-41.42
Domestic services	-37.98
Publishing and printing	-30.99
Infrastructure	-30.76
International organizations	-29.64
Coal mining	-27.02
Building construction	-24.82
Computer and electronic equip.	-24.45
Vehicles and bodies	-23.40

Tabela 20 – Largest and smallest percentage growth in regional formal employment by sector (2016–2021)

Sector	Variation (%)
<i>Largest growth</i>	
Veterinary activities	131.76
IT services	59.31
Aux. financial services and insurance	72.35
Business support services	28.15
Maintenance and repair of machinery	40.29
Engineering and technical testing	42.11
Associative organizations	75.44
Management consultancy	36.48
Advertising and market research	36.77
Education	25.08
<i>Largest declines</i>	
Domestic services	-61.99
International organizations	-50.00
Publishing and printing	-28.85
Creative arts and entertainment	-37.16
Gambling and betting	-31.65
Other transport (except motor vehicles)	-23.44
Extraction of crude petroleum and natural gas	-23.31
Manufacture of tobacco products	-20.20
Radio and TV broadcasting	-15.22
Printing and reproduction	-14.50

Tabela 21 – Allocative Flexibility Index by State

State	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
São Paulo	3.44	5.53	6.35	6.35	8.23	8.41	9.32	7.67	7.68	5.32	3.45	5.18	6.53	10.91
Minas Gerais	4.42	4.87	2.39	6.07	5.58	5.89	7.02	6.62	8.66	6.25	4.69	6.66	4.78	8.49
Espírito Santo	3.95	4.28	4.35	3.18	2.05	4.58	4.80	5.06	3.49	3.37	3.13	4.69	3.62	6.84
Rio Grande do Sul	2.98	4.54	3.25	0.03	1.19	6.08	6.40	4.97	6.49	4.08	4.77	6.12	3.81	6.83
Paraná	3.10	4.44	5.09	4.80	6.75	4.23	7.40	7.84	6.80	7.17	3.22	7.42	2.70	6.61
Goiás	2.84	2.44	2.31	3.39	4.34	5.75	4.05	4.08	2.75	2.48	2.39	3.64	2.18	6.15
Rio de Janeiro	1.04	2.26	3.72	3.55	6.03	5.41	4.07	5.66	4.42	4.05	4.37	5.36	3.93	5.40
Pará	1.41	1.46	1.43	2.43	4.31	4.47	3.63	2.33	3.50	3.63	3.87	2.41	3.32	4.49
Pernambuco	1.49	2.04	0.72	3.07	2.47	2.24	3.76	3.39	4.87	2.91	4.69	4.14	4.18	4.44
Mato Grosso	2.05	1.11	2.29	3.66	3.62	4.71	3.47	2.66	4.44	4.42	4.78	4.70	3.00	4.37
Bahia	1.08	1.64	3.13	4.79	3.03	3.35	4.82	5.43	4.14	6.92	3.70	2.07	3.21	4.29
Sergipe	1.21	0.92	2.03	1.04	1.89	3.38	2.77	2.45	1.67	1.40	1.77	3.11	2.29	2.87
Ceará	1.85	1.53	3.62	4.18	3.63	1.46	3.63	4.32	4.46	3.21	0.91	4.56	2.29	2.86
Amazonas	2.72	0.95	1.71	2.21	2.68	3.09	2.21	2.40	1.24	2.43	3.17	3.20	1.02	2.58
Santa Catarina	3.11	5.12	3.11	5.71	8.53	7.02	1.20	7.37	7.26	5.69	4.46	6.87	5.65	2.35
Distrito Federal	3.04	1.00	1.08	0.92	1.95	3.71	4.48	1.56	4.83	3.19	3.78	2.43	1.95	2.18
Rio Grande do Norte	1.70	1.12	2.24	2.10	1.35	2.30	3.17	2.58	3.82	2.14	3.79	1.54	1.50	2.10
Mato Grosso do Sul	2.46	3.11	1.78	1.90	1.24	4.74	3.86	4.14	1.19	2.90	4.43	2.83	1.63	2.05
Piauí	2.47	1.01	2.49	1.12	1.04	1.66	2.62	1.41	1.37	1.99	1.51	2.30	2.38	2.04
Tocantins	1.65	1.55	1.94	2.12	0.99	2.49	2.42	2.47	1.34	1.95	1.09	1.12	1.33	2.01
Acre	0.87	0.91	1.21	1.01	1.54	0.69	1.70	0.83	0.60	0.88	0.13	3.07	1.15	1.92
Paraíba	2.04	1.94	0.48	1.86	3.04	2.96	4.16	4.13	1.10	2.39	1.22	2.24	2.01	1.57
Rondônia	1.20	1.80	2.61	2.30	2.92	2.23	1.68	0.84	1.35	2.59	3.80	2.27	3.08	1.52
Amapá	0.34	0.06	0.32	1.55	1.00	2.04	1.83	1.46	0.48	0.52	1.37	1.98	0.37	1.52
Alagoas	1.48	0.71	4.21	1.22	1.37	1.66	3.53	1.38	0.79	2.30	3.35	1.85	3.23	1.41
Roraima	0.61	0.12	0.81	1.37	1.63	1.50	0.94	1.08	1.34	1.86	0.73	1.32	1.15	1.29
Maranhão	1.13	1.41	0.49	2.18	2.86	3.16	0.71	3.85	1.26	1.55	1.24	2.59	0.18	1.27

Tabela 22 – Regression Results for the First Period (2010-2016)

	<i>Dependent variable: Allocative Flexibility (2010–2016)</i>				
	(1)	(2)	(3)	(4)	(5)
PPC	0.313*** (0.057)	0.158*** (0.038)	0.099*** (0.031)	0.150** (0.074)	0.099 (0.098)
HH	−0.193*** (0.055)	−0.186*** (0.036)	−0.002 (0.030)	0.067 (0.071)	0.142 (0.094)
N	0.150*** (0.055)	0.035 (0.036)	0.042 (0.029)	−0.032 (0.071)	−0.077 (0.094)
Constant	1.069*** (0.051)	0.718*** (0.034)	0.425*** (0.027)	0.660*** (0.066)	0.662*** (0.088)
Observations	133	133	133	133	127
R ²	0.424	0.376	0.130	0.032	0.027
Adjusted R ²	0.410	0.362	0.110	0.009	0.003
Residual Standard Error	0.585 (df = 129)	0.388 (df = 129)	0.315 (df = 129)	0.757 (df = 129)	0.991 (df = 123)
F Statistic	31.597*** (df = 3; 129)	25.929*** (df = 3; 129)	6.444*** (df = 3; 129)	1.407 (df = 3; 129)	1.126 (df = 3; 123)

Note: *p<0.1; **p<0.05; ***p<0.01

Tabela 23 – Regression Results for the Second Period (2016–2021)

	<i>Dependent variable: Allocative Flexibility (2016–2021)</i>				
	(1)	(2)	(3)	(4)	(5)
PPC	0.409*** (0.076)	0.315*** (0.054)	0.173*** (0.056)	0.103 (0.063)	–0.582 (1.010)
HH	–0.182** (0.074)	–0.080 (0.053)	0.032 (0.054)	0.055 (0.062)	0.647 (0.993)
N	0.286*** (0.073)	0.152*** (0.052)	0.145*** (0.054)	0.026 (0.061)	–0.210 (0.969)
Constant	1.298*** (0.069)	0.834*** (0.049)	0.723*** (0.050)	0.580*** (0.057)	1.855** (0.909)
Observations	133	133	133	133	127
R ²	0.412	0.370	0.158	0.026	0.012
Adjusted R ²	0.399	0.355	0.138	0.003	–0.012
Residual Standard Error	0.793 (df = 129)	0.562 (df = 129)	0.580 (df = 129)	0.659 (df = 129)	10.249 (df = 123)
F Statistic	30.170*** (df = 3; 129)	25.211*** (df = 3; 129)	8.059*** (df = 3; 129)	1.142 (df = 3; 129)	0.487 (df = 3; 123)
<i>Note:</i>				*p<0.1; **p<0.05; ***p<0.01	

Dependent Variable:	ln_wage	
Model:	(1)	(2)
<i>Variables</i>		
ln_unemployment	-0.0544*** (0.0127)	-0.0603*** (0.0201)
age	0.1217*** (0.0029)	0.0607*** (0.0048)
age squared	-0.0012*** (3.63×10^{-5})	-0.0007*** (5.87×10^{-5})
sex2	-0.3023*** (0.0133)	-0.1982*** (0.0099)
race_color2	0.0456* (0.0253)	0.1099*** (0.0233)
race_color4	-0.1336*** (0.0336)	-0.0682** (0.0288)
race_color6	0.1953** (0.0877)	0.2175** (0.0899)
race_color8	-0.0804** (0.0354)	-0.0214 (0.0299)
race_color9	-0.1266*** (0.0296)	-0.0750*** (0.0248)
race_color99	0.1934 (0.1255)	0.4387** (0.1672)
<i>Fixed-effects</i>		
state	Yes	Yes
year	Yes	Yes
<i>Fit statistics</i>		
Observations	35,940,718	292,905,906
R ²	0.39706	0.29158
Within R ²	0.29927	0.10345

Clustered (state) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*